

Supplementary Material for “Revisiting Metaheuristic Comparisons for Continuous Wind Farm Layout Optimization: Baseline Configuration, Component Analysis and a Two-Phase PSO-VNS Design”

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Tables, figures, sections, algorithms and equations of this document are numbered S1, S2, ...; references without the prefix S (e.g., Table II or (4)) point to the main text. The sections follow the order of the main text: details of the evaluation protocol and of the implementation (Section S-I), of the effect of the baseline setting and the main comparison (Section S-II), of the component analysis (Section S-III), of the tests beyond the benchmark (Section S-IV) and of the sensitivity to the modeling assumptions (Section S-V), followed by the model derivations (Section S-VI), a note on published Kusiak–Song results (Section S-VII) and a map from each result to the script that produces it (Section S-VIII). Unless stated otherwise, all results are based on 30 seed-paired runs of 6,030 evaluations (objective calls, including the initial population and, for MS-SLSQP, the finite-difference gradient calls) with random initialization; the method names are those of the main text, and PSO denotes the constriction setting (7) unless the old setting is named. All Horns Rev 1 results use the corrected direction binning of Section S-I-G. All tables of Sections S-II–S-V are generated by `analysis/mpce_results.py` from the published run data.

S-I Evaluation Protocol and Implementation Details

This section gives the data, the pseudocode and the settings needed to reproduce the protocol of Section V: the two wind data sets of the 68 benchmark cases, the salp-swarm methods compared as Phase-1 alternatives in the component analysis, the details of the ranking and the tests (Sections S-I-A–S-I-C), and the implementation details that the main text does not state: iteration counts, MS-SLSQP, the feasibility-preserving initialization, the Horns Rev 1 model and the feasibility tolerance (Sections S-I-D–S-I-H). All settings are those of the code in `analysis/` (Section S-VIII).

S-I-A Wind Data

Table S1 lists the two wind data sets of the Kusiak–Song benchmark [1, 4].

Table S1: Wind Data Sets I and II: probability p_l of direction bin l ($[15(l-1), 15l]^\circ$, direction the wind blows towards, counter-clockwise from east) and Weibull scale ψ ; $\zeta = 2$ in all bins. The most probable bin is $l = 7$ (90° – 105° towards, i.e., wind from 165° – 180° in the meteorological convention $\theta_{\text{met}} = 270^\circ - \theta$) for Data Set I and $l = 13$ (180° – 195° towards, i.e., wind from 75° – 90°) for Data Set II.

Bin l	1	2	3	4	5	6	7	8	9	10	11	12
DS I: p_l ($\psi = 13$)	0.000	0.010	0.010	0.010	0.010	0.200	0.600	0.010	0.010	0.010	0.010	0.010
DS II: ψ	7.0	5.0	5.0	5.0	5.0	4.0	5.0	6.0	7.0	7.0	8.0	9.5
DS II: p_l	0.0002	0.0080	0.0227	0.0242	0.0225	0.0339	0.0423	0.0290	0.0617	0.0813	0.0994	0.1394
Bin l	13	14	15	16	17	18	19	20	21	22	23	24
DS I: p_l ($\psi = 13$)	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.000
DS II: ψ	10.0	8.5	8.5	6.5	4.6	2.6	8.0	5.0	6.4	5.2	4.5	3.9
DS II: p_l	0.1839	0.1115	0.0765	0.0080	0.0051	0.0019	0.0012	0.0010	0.0017	0.0031	0.0097	0.0317

S-I-B Salp Swarm Algorithm and Its Laplacian Variant

Algorithm S1 summarizes SSA [5] and LX-SSA [6] as used in Phase 1 of SSA-VNS and LX-SSA-VNS and as stand-alone methods (Section III-B). A follower moves to the midpoint of its current position and the *new* position of the preceding salp (already updated in the same iteration); in LX-SSA it also forms the Laplace candidate (10), evaluates both candidates and keeps the better one. The new population is then evaluated as a

whole, so a kept LX-SSA follower is evaluated twice; with the N_p candidate evaluations of the $N_p/2$ followers, one LX-SSA iteration therefore costs $2N_p$ evaluations and one SSA iteration N_p . At 6,030 evaluations, stand-alone SSA runs for $T = 200$ and LX-SSA for $T = 100$ iterations; in the hybrids with $\omega = 0.5$, Phase 1 runs for $T = 100$ (SSA) and $T = 50$ (LX-SSA) iterations, i.e., 3,030 evaluations each, and ξ_1 decreases over these T iterations. Both methods use the code of the previous study, as stand-alone methods and inside the hybrids (Section V).

Algorithm S1 SSA and LX-SSA (lines marked LX: LX-SSA only); evaluations per line in brackets

```

1: Draw and evaluate  $N_p$  salps; sort them by  $F_p$  and set the best as food source  $\mathbf{H}$ .  $\triangleright [N_p]$ 
2: for  $t = 1$  to  $T$  do
3:   Update  $\xi_1 = 2 \exp[-(4t/T)^2]$ .
4:   for salp  $i = 1, \dots, N_p$  do
5:     if  $i \leq N_p/2$  then
6:       Update leader  $i$  using (9) (new position  $\tilde{\mathbf{X}}^i$ ).
7:     else
8:       Form the follower candidate  $\mathbf{Z}^i = \frac{1}{2}(\mathbf{X}^i + \tilde{\mathbf{X}}^{i-1})$  and set  $\tilde{\mathbf{X}}^i \leftarrow \mathbf{Z}^i$ .
9:       LX: form  $\mathbf{Y}^i$  using (10), box-repair  $\mathbf{Y}^i$  and  $\mathbf{Z}^i$ , evaluate both and set  $\tilde{\mathbf{X}}^i$  to the better one.  $\triangleright [2$ 
        per follower]
10:    end if
11:  end for
12:  Box-repair  $\tilde{\mathbf{X}}$ , set  $\mathbf{X} \leftarrow \tilde{\mathbf{X}}$ , evaluate  $F_p$  for all salps, sort them, and update  $\mathbf{H}$ .  $\triangleright [N_p]$ 
13: end for
14: return  $\mathbf{H}$ .
```

S-I-C Ranking and Test Details

For the paired run-level tests, a run is scored by its benchmark objective if it is feasible; an infeasible run is scored below every feasible run, and infeasible runs are ordered among themselves by their minimum spacing (larger is better; boundary violations are not used). Paired differences with $|d| \leq 10^{-9}$ are treated as zero and dropped, both in the Wilcoxon signed-rank test and in the rank-biserial correlation r_{rb} . Case means are rounded to 10^{-6} objective units before ranking, so that exact ties (e.g., all methods at the wake-free optimum for $N = 2$) are not broken by floating-point noise. Holm’s correction for the run-level tests is applied within each case over the comparisons of one table; the main-comparison table (Table III) and the ablation table (Table IV) are different families, so the same pair of methods (e.g., PSO-VNS vs. PSO) can have slightly different W/T/L tallies in the two tables. The W/T/L counts are descriptive tallies. In the case-mean Wilcoxon test, a case in which only one of the two methods has at least 15 feasible runs enters as a difference in its favor that is larger than every observed difference, and a case in which neither has is dropped (Section V). The 68 cases are not independent (nested N within six farm and wind combinations), so all p values describe this benchmark, not a population of farms.

S-I-D Iteration Counts and Phase-1 Budgets

With $N_p = 30$ and $B = 6,030$, stand-alone PSO, SSA and DE run for $(B - N_p)/N_p = 200$ iterations and LX-SSA for $(B - N_p)/(2N_p) = 100$. A hybrid with split ω runs $T_1 = \text{round}((\omega B - N_p)/c)$ Phase-1 iterations, where c is the number of evaluations per iteration (N_p for PSO and SSA, $2N_p$ for LX-SSA) and Python’s round-half-to-even rule applies; for $\omega = 0.5$ the argument is 99.5 for PSO and SSA, which is rounded to $T_1 = 100$ (3,030 evaluations), and LX-SSA gets $T_1 = 50$ (3,030 evaluations). For the split study, $\omega = 0.25$ gives 49 PSO iterations (1,500 evaluations) and $\omega = 0.75$ gives 150 (4,530 evaluations). RS-VNS evaluates $\text{round}(\omega B) = 3,015$ random layouts (including the seeded initial population of 30), about the same effort as the 3,030 Phase-1 evaluations of the swarm hybrids; its random layouts, like all random initial layouts, are uniform in the bounding square $[-r, r]^{2N}$, so all N turbines lie inside the circle only with probability $(\pi/4)^N$. DE/rand/1/bin (scale factor 0.5, crossover rate 0.9) clips the mutant to the box before crossover and replaces a member immediately when its trial vector is better (asynchronous one-to-one selection).

S-I-E MS-SLSQP

MS-SLSQP does not minimize F_p : it minimizes the wake loss (without penalty) under explicit inequality constraints, using SciPy’s SLSQP [8] with the bounds $[-r, r]$ on every coordinate, at most 200 iterations per start, $\text{ftol} = 10^{-9}$ and forward-difference gradients with a step of 1 m. The constraints are scaled, $(\ell_{ij}^2 - \ell_{\min}^2)/\ell_{\min}^2 \geq 0$

for every pair and $(r^2 - x_i^2 - y_i^2)/r^2 \geq 0$ for every turbine (for Horns Rev 1, the signed distance of every turbine to every edge of the parallelogram divided by the half-width of the search box, 943 m). The 30 initial layouts are evaluated with F_p and used as starts in increasing order of F_p ; after the 30 initial layouts, every further start is a new random layout drawn from the same generator (uniform in the box, or feasible under the feasibility-preserving initialization), until the budget is used up. Every objective call, including the finite-difference calls, counts as an evaluation. MS-SLSQP returns the best evaluated layout (by wake loss) whose scaled constraints are all $\geq -10^{-9}$; the reported feasibility of this layout is then judged with the common tolerance of 10^{-6} m (Section V). These are standard, untuned settings; since the top-hat Jensen wake makes the wake loss discontinuous in the coordinates, finite-difference gradients are unreliable, which handicaps MS-SLSQP on this benchmark.

S-I-F Feasibility-Preserving Initialization

Algorithm S2 generates each of the N_p initial layouts of the feasible-start study (Section VIII-A); it uses no objective evaluations, so it does not count toward the budget. With spacing $s = \ell_{\min}$, it minimizes the smooth packing penalty

$$P(\mathbf{z}) = \frac{1}{\tilde{s}^4} \left[\frac{1}{2} \sum_{i < j} \max(0, \tilde{s}^2 - \ell_{ij}^2)^2 + \sum_i b_i \right], \quad \tilde{s} = 1.0005 s, \quad (\text{S1})$$

with $b_i = \max(0, x_i^2 + y_i^2 - \tilde{r}^2)^2$, $\tilde{r} = (1 - 10^{-7}) r$, for the circular farms and $b_i = \tilde{s}^2 \sum_e \max(0, \delta_{ie})^2$, with δ_{ie} the signed distance of turbine i outside edge e , for the Horns Rev 1 parallelogram. The slightly inflated spacing and shrunken radius leave a margin, so that the result passes the feasibility test. All random draws come from the seeded global generator, so the runs of all methods with the same seed start from the same feasible initial population and stay paired; under this initialization the later random restarts of MS-SLSQP also use Algorithm S2. RS-VNS was not run with this initialization, since it would need one packing solve for each of its 3,015 random layouts.

Algorithm S2 Feasibility-preserving initialization of one layout (N turbines, spacing s , circle of radius r or convex polygon)

- 1: **for** try = 1 to 200 **do**
 - 2: Draw N positions uniformly in the site (circle: uniform in the disc; polygon: rejection sampling from its bounding box).
 - 3: Minimize P (S1) from these positions with L-BFGS-B (analytic gradient; at most 3,000 iterations, $\text{ftol} = 10^{-15}$, $\text{gtol} = 10^{-12}$).
 - 4: Circle: move every turbine that lies outside the circle radially back onto it.
 - 5: **if** every pairwise distance is $\geq s - 10^{-6}$ m and every turbine lies within 10^{-6} m of the site **then**
 - 6: **return** the layout.
 - 7: **end if**
 - 8: **end for**
 - 9: Stop with an error.
-

S-I-G Horns Rev 1 Model

The Horns Rev 1 block (Section VIII-B) uses the V80 power and thrust curves and the 12-sector Weibull wind climate of PyWake’s Horns Rev 1 example [9], with the sector frequencies normalized to one. The boundary is the parallelogram through the four corner turbines of the installed 16-turbine block, enlarged by 0.1% about the centre of the block so that the installed turbines (integer coordinates) lie inside it; in F_p (4), the boundary violation of a turbine is its distance (m) outside the parallelogram, and the spacing violation is $\ell_{\min} - \ell_{ij}$ (m) with $\ell_{\min} = 4D = 320$ m. The search box is the square of half-width 943 m (the largest absolute coordinate of the parallelogram) centred on the block; the same half-width is the r of the VNS radii. The wake model is the Jensen top-hat wake with $K = 0.04$, a hub-centre in-wake test (turbine i is waked by j if it lies downstream of j and within $R + Kx$ of its axis, where x is the downstream distance; there is no rotor-overlap fraction), the thrust coefficient at the free-stream speed and root-sum-square superposition. The wind speed is discretized in 1-m/s bins centred at 3, 4, $\dots$, 25 m/s, each with the Weibull probability of its ± 0.5 m/s interval. The wind direction is discretized in 5° bins centred at 2.5° , 7.5° , $\dots$, 357.5° (meteorological convention); every 30° sector (centred at 0° , 30° , $\dots$, 330°) contains exactly six bins, each carrying the Weibull parameters of its sector and one sixth of its frequency.

Corrected direction binning. An earlier version of the model centred the bins at 0° , 5° , $\dots$, 355° and assigned them to sectors by rounding half to even, so that the sectors received seven and five bins alternately and the frequencies summed to 1.003. This was corrected as described above, and all Horns Rev 1 runs of this paper

(all methods, budgets and initializations) were repeated with the corrected model; the installed and wake-free AEP values and all Horns Rev 1 results in the main text and in this document come from the corrected model. The check of the model against PyWake is described in Section S-IV-B.

S-I-H Feasibility Tolerance

The penalty F_p treats any positive violation as infeasible, whereas every reported feasibility (final layouts, convergence curves, ranking rule) uses the tolerance of 10^{-6} m on both the spacing and the boundary, in all studies. Because the penalty term is $(1 + \mu g)^2$ with $\mu = 10^{10}$, the comparison of F_p values is a feasibility-first rule only for violations above about 10^{-7} : a smaller violation adds only about one unit to F_p and can beat a feasible layout. Conversely, violations between about 10^{-7} and 10^{-6} m are heavily penalized in F_p but counted as feasible in the reporting.

S-II Baseline Setting and Main Comparison: Details

This section documents the runs behind the effect of the baseline setting (Section VI-A) and gives the per-case results, the full case-level statistics, the coordinates of the best layouts and further figures of the main comparison of the eight methods (Section VI-B).

S-II-A Old and Corrected PSO Setting

Table I ranks the method pool of the previous version of this study (LX-SSA-VNS, SSA-VNS, LX-SSA, SSA, PSO, DE, VNS and MS-SLSQP) twice: once with PSO in the *old setting* $w = 0.7$, $c_1 = c_2 = 2$, and once with the *corrected* constriction setting (7) used for PSO everywhere else in this paper. The old setting satisfies the order-1 (mean) stability condition but violates the order-2 (variance) condition (8) of [7] ($c_1 + c_2 = 4$ against the bound 3.50 for $w = 0.7$), so the particle positions do not converge in variance. Both conditions are derived under stagnation and for an unbounded search space; in our code the velocities are kept when a position is clipped to the box, which, together with the large variance, plausibly explains the lower feasibility of the old setting (feasible in 86.0% of its runs; Section VI-A). The old-setting runs come from the PSO code of the previous study (`analysis/authors_optimizers.py`, run by `analysis/full_grid_experiments.py`); the corrected runs use the same code, in which only the three coefficients are changed. Both settings thus share the population of 30, the 200 iterations, the zero initial velocities, the box repair and the immediate update of the personal and global bests, and both are run on the same 68 cases with the same penalized objective (4), the same budget of 6,030 evaluations and the same seeds 1–30. Because every optimizer draws its initial population first after seeding, a run with the old setting starts from the same initial population as the run with the corrected setting and as the runs of all other methods with the same seed, so the comparison of the two settings is seed-paired. The two columns of Table I share the runs of all methods except PSO and use the same feasibility-aware ranking rule (Table II); every change in the ranking therefore comes from the PSO setting alone. The old setting was run only on the 68 benchmark cases at 6,030 evaluations, so the change in the ranking is shown at this budget only. At larger budgets the ranking changes: at 120,030 evaluations SSA-VNS has a better average rank on the six largest cases than PSO (4.17 vs. 5.83; Section VIII-A). All other results of the paper use the corrected setting. Table S2 gives the per-case comparison of the two settings.

S-II-B Per-Case Results

Tables S3–S8 give the mean and standard deviation of the benchmark objective of every method in every case of the main comparison.

Table S2: Per-case ranks (1 = best of 8; feasibility-aware rule) of PSO, SSA-VNS and LX-SSA-VNS in the method pool of Table I with the old PSO setting ($w = 0.7$, $c_1 = c_2 = 2$) and with the constriction setting (68 cases, 6,030 evaluations).

DS	r	N	Old setting			Constriction		
			PSO	SSA-VNS	LX-SSA-VNS	PSO	SSA-VNS	LX-SSA-VNS
I	500	2	4	4	4	4	4	4
I	500	3	4	3	5	3	4	5
I	500	4	4	2	3	1	3	4
I	500	5	1	5	2	1	5	2
I	500	6	2	1	4	1	2	4
I	500	7	3	6	5	1	6	5
I	500	8	6	3	5	1	4	6
I	500	9	7	3	4	1	4	5
I	500	10	8	2	3	2	3	4
I	750	2	4	4	4	4	4	4
I	750	3	4	3	2	3	4	2
I	750	4	4	3	2	1	4	3
I	750	5	3	1	2	1	2	3
I	750	6	3	1	2	1	2	3
I	750	7	4	1	2	1	2	3
I	750	8	4	1	3	1	2	4
I	750	9	7	1	2	1	2	3
I	750	10	7	1	4	1	2	5
I	750	11	7	2	1	1	3	2
I	750	12	7	3	2	1	4	3
I	1000	2	4	4	4	4	4	4
I	1000	3	3	3	3	3	3	3
I	1000	4	4	1	2	1	2	3
I	1000	5	5	1	2	1	2	3
I	1000	6	4	1	2	1	2	3
I	1000	7	4	1	2	1	2	3
I	1000	8	5	1	2	1	2	3
I	1000	9	6	1	2	1	2	3
I	1000	10	6	1	2	1	2	3
I	1000	11	7	1	3	1	2	4
I	1000	12	7	1	3	1	2	4
I	1000	13	7	1	2	1	2	3
I	1000	14	7	2	1	1	3	2
I	1000	15	7	1	2	1	2	3
II	500	2	4	4	4	4	4	4
II	500	3	4	3	2	4	3	2
II	500	4	1	3	4	1	3	4
II	500	5	2	3	1	1	3	2
II	500	6	4	1	2	1	2	3
II	500	7	6	2	4	1	3	5
II	500	8	7	2	3	1	3	4
II	500	9	7	2	3	1	3	4
II	500	10	8	2	4	1	3	5
II	750	2	4	4	4	4	4	4
II	750	3	4	1.5	3	4	1.5	3
II	750	4	3	4	2	3	4	2
II	750	5	3	1	2	1	2	3
II	750	6	3	1	2	1	2	3
II	750	7	4	1	2	1	2	3
II	750	8	6	1	2	1	2	3
II	750	9	7	1	2	1	2	3
II	750	10	7	2	4	1	3	5
II	750	11	6	2	4	1	3	5
II	750	12	7	3	2	1	4	3
II	1000	2	4	4	4	4	4	4
II	1000	3	5	2.5	2.5	3	3	3
II	1000	4	4	2	3	2	3	4
II	1000	5	4	2	1	1	3	2
II	1000	6	4	1	2	1	2	3
II	1000	7	3	2	1	1	3	2
II	1000	8	5	1	2	1	2	3
II	1000	9	7	1	2	1	2	3
II	1000	10	7	1	2	1	2	3
II	1000	11	7	1	3	1	2	4
II	1000	12	7	1	3	1	2	4
II	1000	13	7	1	2	1	2	3
II	1000	14	6	1	2	1	2	3
II	1000	15	7	1	2	1	2	3

Table S3: Data Set I, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Wake-free	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28086.7 (4.5)
3	42137.2	42131.3 (8.5)	42128.7 (8.8)	42128.5 (10.7)	42114.9 (10.3)	42114.8 (14.2)	42130.7 (8.8)	42131.6 (9.7)	42119.9 (11.0)
4	56182.9	56114.4 (20.7)	56110.0 (25.6)	56101.8 (25.3)	56086.8 (19.8)	56078.8 (22.6)	56092.8 (29.8)	56106.1 (21.7)	56057.0 (55.8)
5	70228.7	70051.2 (28.5)	70053.1 (29.3)	69967.2 (159.2)	69994.0 (34.0)	69972.6 (72.0)	69960.5 (78.6)	69876.9 (199.8)	69876.4 (139.3)
6	84274.4	83918.5 (66.9)	83925.6 (55.8)	83780.8 (213.2)	83535.6 (372.5)	83544.9 (465.3)	83326.7 (761.5)	83299.3 (524.5)	83636.4 (227.9)
7	98320.2	97588.7 (292.3)	97569.6 (325.2)	96529.9 (794.9)	96757.0 (692.4)	96380.6 (1013.8)	95903.3 (1316.3) ¹⁵	96967.1 (590.6)	97049.0 (477.4)
8	112365.9	110777.2 (718.5)	110957.4 (615.4)	109316.4 (1117.4)	109172.6 (1362.4)	108258.3 (1704.3) ²⁸	107632.7 (1592.8) ²	109493.5 (2035.2)	110233.3 (571.9)
9	126411.6	123153.2 (1132.3)	123499.5 (1006.4)	121985.3 (1201.8)	120043.1 (2755.8) ²⁵	119134.2 (2666.4) ²¹	– ⁰	122259.2 (1232.0)	123307.1 (814.5)
10	140457.4	135160.5 (1708.4)	135396.0 (1790.4)	134321.3 (1482.4) ²⁶	132616.3 (2328.0) ¹²	130967.4 (3191.0) ¹³	130994.8 (0.0) ¹	133567.5 (1820.9)	135642.4 (920.1)

Table S4: Data Set I, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Wake-free	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.0 (1.6)
3	42137.2	42137.2 (0.0)	42137.0 (1.0)	42137.0 (1.0)	42134.4 (4.1)	42134.2 (4.8)	42136.8 (1.5)	42137.2 (0.0)	42133.3 (5.0)
4	56182.9	56170.9 (10.1)	56170.9 (11.3)	56162.3 (13.6)	56153.8 (17.9)	56149.9 (14.0)	56146.9 (17.5)	56167.0 (13.8)	56139.3 (11.7)
5	70228.7	70169.6 (23.7)	70164.8 (25.1)	70150.9 (25.4)	70122.2 (26.3)	70118.1 (28.6)	70083.9 (31.7)	70125.8 (47.0)	70091.5 (35.7)
6	84274.4	84139.4 (24.3)	84130.1 (30.7)	84095.6 (36.9)	84035.2 (58.4)	84033.5 (58.3)	83948.2 (74.1)	84029.2 (141.3)	83965.7 (117.2)
7	98320.2	98064.7 (54.7)	98059.9 (46.4)	97910.8 (164.8)	97875.2 (169.3)	97840.0 (167.7)	97479.1 (285.1)	97774.4 (232.6)	97663.0 (196.3)
8	112365.9	111921.0 (119.4)	111924.1 (87.8)	111683.9 (282.2)	111506.5 (334.5)	111374.0 (460.8)	110595.8 (445.6)	111411.4 (493.5)	111287.8 (290.8)
9	126411.6	125552.7 (330.8)	125639.1 (260.7)	125207.2 (539.4)	124756.2 (610.5)	124698.0 (960.4)	121674.9 (2276.5) ²³	124780.0 (716.8)	124787.6 (424.2)
10	140457.4	139122.6 (345.7)	139212.9 (276.9)	138304.3 (685.9)	137817.6 (1160.1)	137748.6 (1116.3)	134937.7 (998.5) ⁴	138170.4 (764.7)	138182.1 (569.0)
11	154503.1	152343.5 (565.1)	152626.6 (443.6)	151286.3 (1117.4)	150404.1 (1042.2) ²⁹	149863.8 (1886.6) ²⁹	148555.7 (1134.8) ³	151004.2 (1339.0)	151173.5 (598.0)
12	168548.8	165526.4 (774.4)	165826.3 (766.5)	164279.2 (1284.1)	162903.5 (1685.8)	161630.5 (1988.2) ²⁸	– ⁰	163712.4 (1575.6)	164478.5 (634.3)

Table S5: Data Set I, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Wake-free	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.3 (0.7)
3	42137.2	42137.2 (0.0)	42137.2 (0.0)	42137.2 (0.0)	42136.7 (1.6)	42136.7 (1.4)	42137.2 (0.0)	42137.2 (0.0)	42136.6 (1.8)
4	56182.9	56181.4 (3.1)	56181.8 (3.1)	56180.0 (3.8)	56169.4 (9.2)	56172.9 (7.8)	56166.9 (8.4)	56177.3 (8.6)	56165.3 (10.2)
5	70228.7	70209.8 (10.8)	70208.0 (13.1)	70202.1 (15.9)	70174.0 (19.7)	70183.3 (21.0)	70148.7 (22.4)	70193.3 (24.5)	70159.6 (39.9)
6	84274.4	84214.5 (19.7)	84213.1 (24.2)	84175.9 (36.0)	84149.9 (29.7)	84142.7 (23.5)	84096.1 (24.0)	84170.0 (49.5)	84064.6 (113.6)
7	98320.2	98192.1 (32.2)	98178.1 (38.0)	98128.4 (59.9)	98084.7 (47.0)	98078.4 (82.1)	97942.3 (67.9)	98064.6 (118.7)	97930.4 (157.1)
8	112365.9	112142.9 (48.9)	112125.1 (48.4)	112008.3 (131.2)	111947.1 (125.7)	111924.6 (164.7)	111608.3 (168.9)	111862.5 (246.1)	111736.9 (163.4)
9	126411.6	126003.5 (103.1)	125993.3 (89.8)	125842.0 (277.4)	125761.1 (168.4)	125701.5 (265.4)	124996.3 (282.9)	125589.7 (312.6)	125450.0 (305.1)
10	140457.4	139874.2 (102.1)	139834.3 (108.2)	139561.7 (359.4)	139348.0 (433.0)	139354.2 (388.5)	137969.0 (623.0)	139290.3 (447.7)	138991.7 (381.3)
11	154503.1	153621.0 (192.7)	153621.2 (168.9)	153184.6 (292.3)	152825.1 (385.8)	152736.8 (558.8)	150189.7 (1068.9) ²⁹	153010.5 (393.8)	152538.4 (405.6)
12	168548.8	167262.1 (383.9)	167223.9 (330.8)	166530.1 (760.0)	166091.9 (816.6)	165814.4 (827.9)	162390.4 (1765.9) ¹⁷	166510.6 (669.8)	166171.0 (462.2)
13	182594.6	180623.7 (496.4)	180638.4 (495.0)	179973.5 (642.1)	179101.5 (1028.9)	178913.6 (967.4)	174763.4 (1389.6) ⁴	179762.6 (1091.2)	179275.1 (644.9)
14	196640.3	194092.5 (813.0)	194082.6 (688.3)	193260.8 (939.8)	192235.7 (1504.8)	191046.6 (1754.6)	– ⁰	192874.6 (1745.6)	192657.7 (605.3)
15	210686.1	207308.1 (894.1)	207263.9 (720.0)	207119.3 (639.3)	204777.6 (1420.5)	203681.8 (1899.5) ²⁹	– ⁰	206486.7 (1093.6)	205876.8 (798.9)

Table S6: Data Set II, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Wake-free	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14626.0 (5.6)
3	21946.1	21909.3 (15.9)	21904.5 (13.8)	21907.1 (18.9)	21900.9 (11.1)	21899.7 (9.5)	21901.2 (5.5)	21932.3 (17.2)	21892.0 (45.9)
4	29261.5	29124.8 (63.8)	29142.9 (54.2)	29067.5 (60.7)	29028.7 (93.8)	29004.2 (87.4)	29044.2 (61.5)	29070.8 (94.1)	28993.5 (96.1)
5	36576.9	36100.0 (100.0)	36118.0 (117.9)	35958.0 (130.6)	35918.9 (74.0)	35880.1 (168.6)	35789.6 (193.8)	35878.2 (160.9)	35914.9 (144.5)
6	43892.3	42851.5 (166.9)	42857.9 (163.5)	42638.7 (198.9)	42444.3 (224.5)	42512.1 (227.9)	42257.5 (374.9)	42560.3 (202.5)	42601.8 (178.6)
7	51207.6	49332.5 (195.5)	49337.3 (189.8)	49008.6 (239.7)	48887.9 (304.1)	48626.5 (340.5)	48203.9 (767.8) ¹⁵	49032.1 (314.4)	48959.7 (202.6)
8	58523.0	55727.9 (300.0)	55668.1 (312.6)	55274.9 (503.7)	54900.5 (471.3)	54699.5 (620.7) ²⁸	54451.3 (618.7) ²	55182.8 (738.6)	55427.1 (303.9)
9	65838.4	61754.3 (344.3)	61643.9 (481.7)	61380.0 (541.4)	60581.0 (844.8) ²⁵	59963.3 (652.2) ²¹	– ⁰	61166.8 (574.8)	61516.6 (326.2)
10	73153.8	67773.5 (477.6)	67472.2 (637.3)	67106.6 (779.5) ²⁵	65880.7 (761.6) ¹²	65868.7 (586.0) ¹³	64240.2 (0.0) ¹	67056.0 (749.1)	67297.9 (402.5) ²⁹

Table S7: Data Set II, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Wake-free	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.2 (1.8)
3	21946.1	21945.5 (2.5)	21945.0 (3.9)	21946.1 (0.0)	21940.0 (8.2)	21937.3 (8.1)	21942.5 (5.8)	21946.1 (0.0)	21938.3 (11.3)
4	29261.5	29225.9 (22.5)	29221.2 (22.7)	29210.5 (34.5)	29182.5 (36.2)	29201.4 (25.7)	29195.3 (26.2)	29227.3 (29.9)	29175.2 (48.2)
5	36576.9	36473.4 (36.9)	36480.6 (37.9)	36417.1 (67.3)	36332.0 (91.4)	36366.8 (74.3)	36277.7 (105.9)	36389.7 (97.1)	36267.4 (108.2)
6	43892.3	43596.3 (87.7)	43611.5 (96.3)	43500.1 (119.6)	43419.5 (153.1)	43377.4 (175.7)	43142.2 (144.8)	43367.4 (201.2)	43264.0 (140.5)
7	51207.6	50584.2 (207.2)	50592.0 (149.5)	50370.1 (169.6)	50264.8 (202.5)	50095.4 (193.6)	49696.6 (156.3)	50180.2 (303.8)	50174.3 (220.0)
8	58523.0	57411.1 (173.3)	57404.8 (185.9)	57033.9 (298.6)	56806.2 (303.6)	56659.2 (332.0)	56033.1 (331.4)	56855.9 (391.5)	56837.8 (340.9)
9	65838.4	64016.2 (259.2)	63991.1 (213.9)	63632.1 (366.8)	63291.6 (402.5)	63252.7 (342.8)	61970.8 (763.0) ²³	63217.0 (515.4)	63472.4 (272.8)
10	73153.8	70558.3 (205.2)	70484.8 (208.8)	70000.5 (393.6)	69520.3 (512.6)	69480.7 (385.7)	68103.2 (423.7) ⁴	69930.2 (355.0)	70128.5 (305.6)
11	80469.2	76912.3 (421.8)	76804.4 (395.8)	76265.0 (512.1)	75742.5 (677.1) ²⁹	75181.6 (681.1) ²⁹	74596.6 (476.3) ³	76099.7 (566.3)	76420.9 (338.9)
12	87784.5	82968.9 (469.9)	82874.2 (531.0)	82500.8 (492.0)	81822.1 (625.7)	81341.8 (792.1) ²⁸	– ⁰	82368.6 (603.1)	82792.2 (462.3)

Table S8: Data Set II, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations. Bold: highest mean among the methods with at least 15 feasible runs. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

N	Wake-free	PSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.4 (1.2)
3	21946.1	21946.1 (0.0)	21946.1 (0.0)	21946.1 (0.0)	21945.6 (1.8)	21945.3 (2.8)	21946.1 (0.0)	21946.1 (0.0)	21943.9 (3.7)
4	29261.5	29254.5 (7.9)	29251.3 (10.2)	29247.6 (12.2)	29232.5 (18.5)	29228.0 (14.9)	29225.9 (12.1)	29253.1 (11.7)	29220.8 (35.1)
5	36576.9	36514.3 (28.7)	36519.7 (32.4)	36510.6 (35.3)	36485.9 (37.5)	36468.2 (46.1)	36402.8 (39.4)	36498.4 (49.1)	36426.0 (58.3)
6	43892.3	43776.8 (48.3)	43789.7 (42.7)	43736.8 (52.7)	43626.4 (74.5)	43656.6 (73.3)	43488.5 (69.7)	43686.1 (101.2)	43554.1 (85.6)
7	51207.6	50952.8 (83.2)	50969.1 (79.2)	50800.0 (120.6)	50731.8 (118.8)	50727.3 (119.3)	50442.7 (136.0)	50696.0 (203.8)	50596.6 (137.4)
8	58523.0	58050.4 (138.4)	58051.3 (154.0)	57883.4 (180.0)	57739.3 (142.0)	57645.1 (186.6)	57210.0 (191.8)	57588.6 (311.3)	57528.0 (226.3)
9	65838.4	65052.9 (170.2)	65065.3 (177.0)	64730.4 (233.0)	64476.4 (203.0)	64428.7 (273.1)	63790.1 (227.2)	64430.8 (311.5)	64357.7 (268.5)
10	73153.8	71906.7 (236.9)	71902.4 (238.0)	71541.9 (328.8)	71139.4 (336.8)	71099.4 (266.9)	70190.1 (286.9)	71342.3 (339.4)	71142.5 (257.2)
11	80469.2	78633.2 (289.7)	78478.9 (290.3)	78133.1 (465.6)	77697.5 (425.8)	77561.6 (498.0)	75960.9 (577.6) ²⁹	77948.1 (483.3)	77762.2 (436.7)
12	87784.5	85357.5 (330.7)	85172.5 (311.4)	84647.1 (650.8)	84049.6 (660.3)	83981.2 (515.0)	82211.6 (905.2) ¹⁷	84485.7 (606.8)	84385.5 (437.1)
13	95099.9	91904.0 (428.6)	91685.0 (453.5)	91179.4 (556.4)	90267.0 (482.6)	90303.5 (674.2)	88665.1 (496.0) ⁴	90787.8 (806.9)	90948.3 (429.2)
14	102415.3	98223.0 (410.2)	97889.6 (465.1)	97590.4 (615.0)	96936.5 (669.3)	96229.6 (1019.3)	– ⁰	97079.1 (980.3)	97371.5 (411.6)
15	109730.7	104794.8 (531.9)	104375.6 (501.6)	104035.3 (475.4)	102746.3 (827.8)	102327.4 (838.7) ²⁹	– ⁰	103777.6 (794.4)	103901.2 (406.2)

S-II-C Case-Level Statistics and Best Layouts

Table S9 extends Table II by the full case-level statistics, and Table S10 lists the coordinates of the best layouts for the largest N of each farm (shown in Fig. S8); when PSO-VNS is itself the best method of a farm, Table S10 has a single entry for it.

Table S9: Case-level analysis over the 68 benchmark cases. Ranks of the mean feasible benchmark objective within each case (1 = best); a method with fewer than 15 feasible runs out of 30 in a case is ranked below all other methods, by its number of feasible runs (“<15”: number of such cases). Friedman $\chi_F^2 = 319.3$ (7 d.f.), $p = 4.56 \times 10^{-65}$; Iman–Davenport $F_F = 136.5$, $p = 6.71 \times 10^{-109}$. p_z : Holm-adjusted p of the average-rank z test against PSO-VNS (best-ranked method: PSO-VNS). Because mean-rank post hoc tests depend on the pool of compared methods [11], pairwise two-sided Wilcoxon signed-rank tests on the per-case mean wake losses (%) are also given: p_W (Holm-adjusted over the 7 comparisons), matched-pairs rank-biserial correlation r_{rb} (positive = PSO-VNS better) and mean difference $\overline{\Delta L}$ of the wake loss (percentage points, PSO-VNS minus method, over the cases where both methods have at least 15 feasible runs). “Best”: cases in which the method alone ranks first; “Feas.”: percentage of feasible runs.

Method	Avg. rank	Best	Feas. (%)	<15	p_z	p_W	r_{rb}	$\overline{\Delta L}$ (pp)
PSO-VNS	1.82	29	100.0	0	–	–	–	–
PSO	1.97	25	100.0	0	0.713	0.296	+0.16	–0.018
SSA-VNS	3.45	0	99.6	0	2.04×10^{-4}	5.69×10^{-11}	+1.00	–0.330
VNS	4.40	3	100.0	0	2.42×10^{-9}	1.26×10^{-10}	+0.98	–0.465
SSA	5.38	0	97.6	2	8.32×10^{-17}	4.55×10^{-11}	+1.00	–0.611
MS-SLSQP	5.88	1	100.0	0	1.84×10^{-21}	4.18×10^{-11}	+0.96	–0.435
LX-SSA	6.06	0	96.9	2	3.33×10^{-23}	4.55×10^{-11}	+1.00	–0.765
DE	7.04	0	71.4	18	1.04×10^{-34}	5.59×10^{-11}	+1.00	–0.887

Table S10: Coordinates (m, farm center at the origin) of the best layout of all methods and of the best PSO-VNS layout for the largest N of each farm (6,030 evaluations); objective in benchmark units.

DS	r (m)	N	Method	Objective	Coordinates (x_i, y_i)
I	500	10	MS-SLSQP	138291.2	(469.2, 168.9); (127.1, -90.8); (-183.2, 21.4); (-365.2, -324.6); (-493.9, 32.0); (-320.1, 380.7); (273.4, 411.2); (-38.6, -356.1); (270.8, -412.5); (-6.8, 274.6); (485.3, -32.1); (-346.0, 302.7); (322.3, -382.2); (-376.7, -328.3); (317.9, 385.7); (142.8, 131.7); (-59.3, -101.1); (-46.5, 497.1); (-497.9, -0.5); (-88.7, -476.6); (-71.5, 110.7); (192.8, -653.4); (741.7, 104.8); (395.5, 126.3); (65.5, -302.0); (-728.9, 172.1); (185.8, 655.0); (-226.9, 453.5); (-531.3, -385.6); (534.1, 524.5); (542.8, -343.4); (-223.2, -715.4)
I	500	10	PSO-VNS	137994.8	(-466.8, -48.4); (923.5, 23.2); (684.6, 693.4); (-230.6, 303.0); (524.3, -694.5); (137.7, 67.9); (517.2, 119.0); (137.7, 964.0); (153.5, -960.7); (-232.5, -607.3); (675.4, -366.5); (-684.6, -361.8); (-980.6, 75.4); (-48.5, -210.1); (-687.0, 528.7); (594.1, 135.8); (-268.3, -387.2); (786.8, -517.5); (-79.4, 263.7); (960.7, 115.7); (-852.8, -274.3); (-998.2, 26.9); (104.5, -420.5); (-552.9, -78.4); (342.1, -881.7); (97.7, 888.8); (-84.5, -912.6); (-515.9, 610.4); (-707.3, 234.6); (355.1, 465.8); (499.3, -17.9); (-357.0, 305.5); (302.3, -322.3); (-362.6, -295.2); (336.9, 352.5); (46.0, 150.6); (-45.5, -150.2); (-65.8, 495.6); (-499.5, -2.6); (-102.9, -475.6); (-159.3, -214.1); (492.9, 564.5); (-391.5, 639.1); (121.1, -739.2); (-44.2, 619.1); (-662.7, 13.0); (-641.7, -309.1); (-465.3, 325.3); (-348.1, -575.4); (665.0, -320.3); (98.2, 250.2); (724.3, 15.3)
I	750	12	PSO-VNS	167040.0	(-19.0, 118.7); (-84.9, -328.5); (-602.6, 187.3); (639.7, -523.0); (-942.8, -326.2); (216.5, -974.4); (928.9, -325.3); (-386.7, 906.8); (349.5, 906.0); (-310.8, 528.7); (993.7, 2.3); (322.4, -676.0); (469.3, 84.1); (701.3, 536.4); (-435.2, -798.8)
I	1000	15	PSO	208908.8	
I	1000	15	PSO-VNS	208720.0	
II	500	10	PSO-VNS	68621.3	
II	750	12	PSO-VNS	83996.8	
II	1000	15	PSO-VNS	105675.2	

S-II-D Additional Figures

Fig. S1 shows the average ranks of Table II and Fig. S2 the mean wake loss versus the number of turbines. Figs. S3–S8 show the convergence, the distribution of the final wake loss, the feasibility and the best layouts of the eight methods of the main comparison.

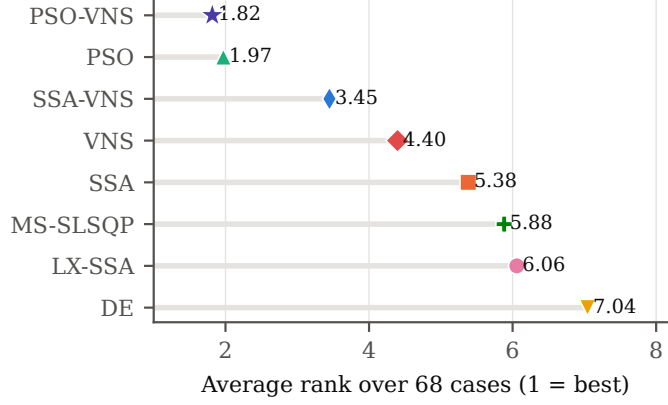


Figure S1: Average rank of the eight methods of the main comparison over the 68 benchmark cases (feasibility-aware ranking of Table II; 1 = best).

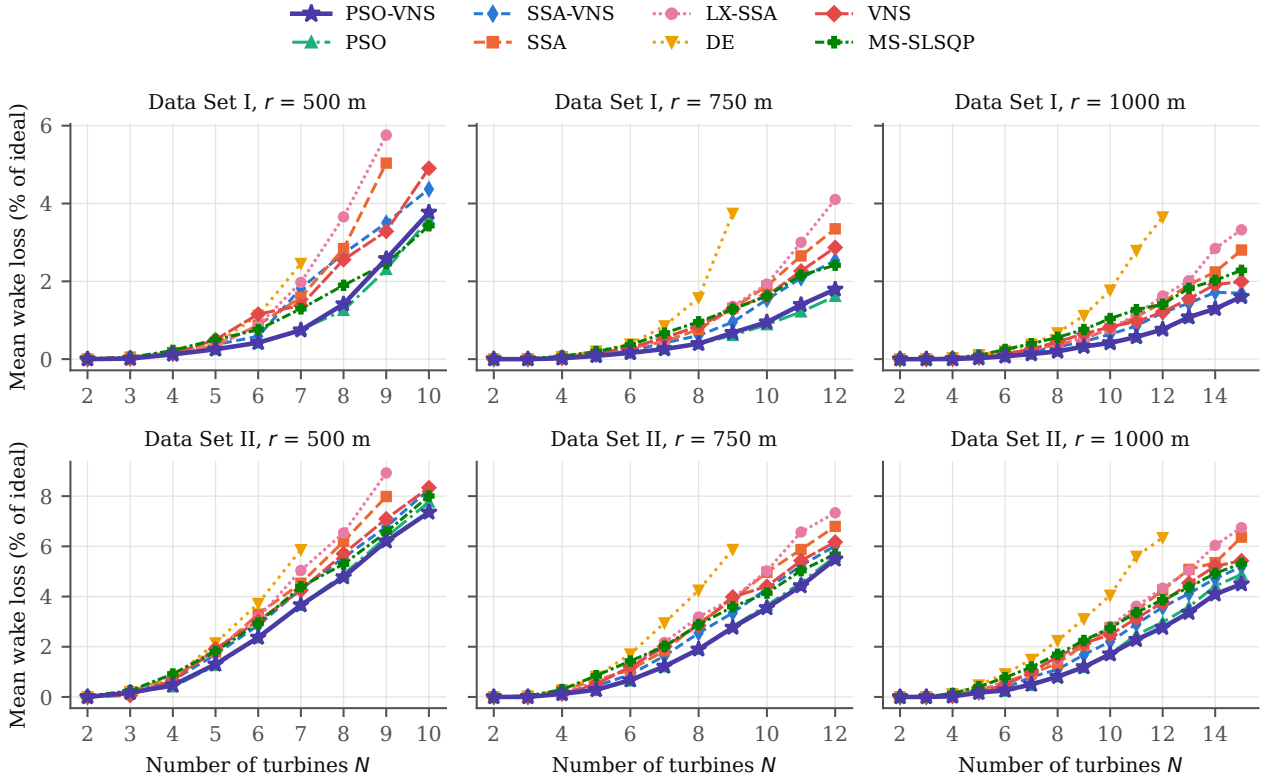


Figure S2: Mean wake loss (% of the wake-free benchmark objective) of the feasible runs versus the prescribed number of turbines N , for each farm and wind data set (30 runs of 6,030 evaluations, random initialization). A point is drawn only when the method has at least 15 feasible runs in that case, so a curve (mostly DE) stops where the method has too few feasible runs.

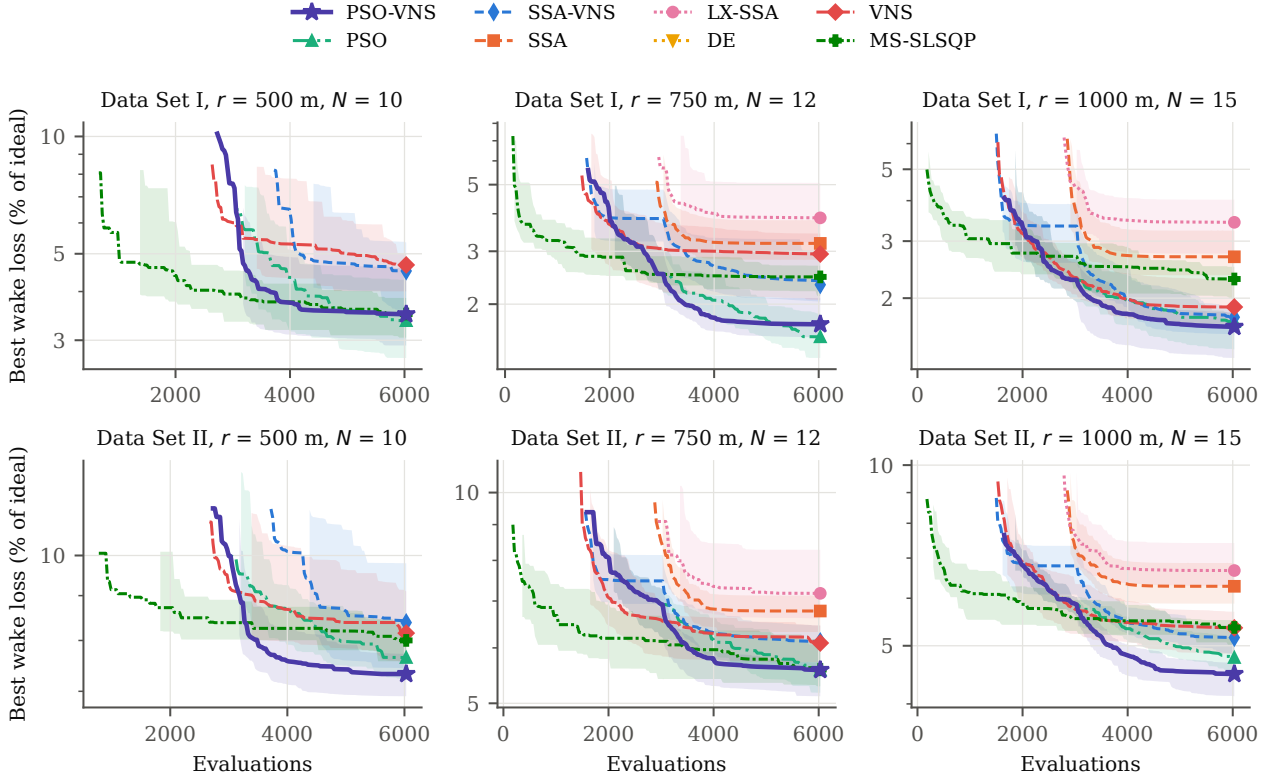


Figure S3: Convergence of the eight methods of the main comparison for the largest N of each farm: median (line) and interquartile range (band) of the best feasible wake loss over 30 runs versus the number of evaluations (objective-function calls), logarithmic vertical axis. The median is drawn once at least 15 of the 30 runs have found a feasible layout and the upper quartile once at least 23 have, so curves start late where early layouts are infeasible; a method that never reaches this share (DE in the largest cases) has no curve in the panel.

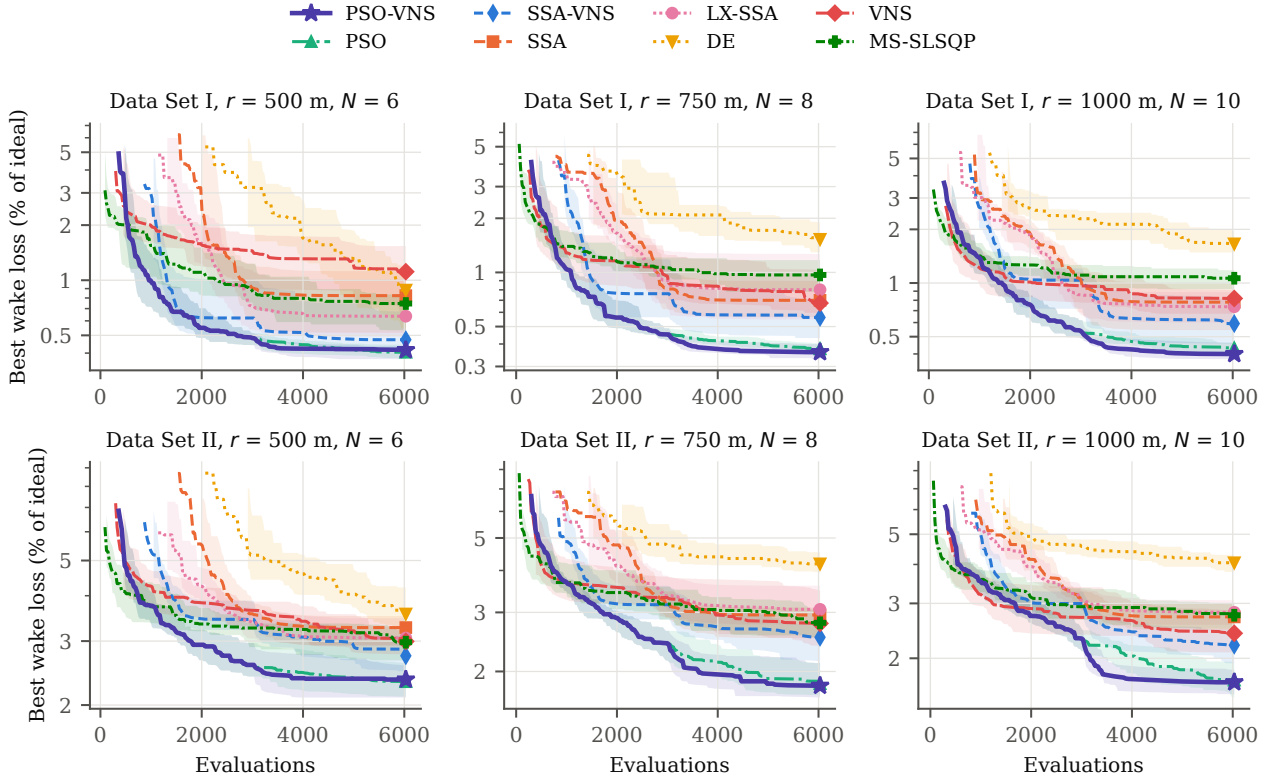


Figure S4: As Fig. S3, for the middle N of each farm ($N = 6, 8$ and 10 for $r = 500, 750$ and 1000 m).

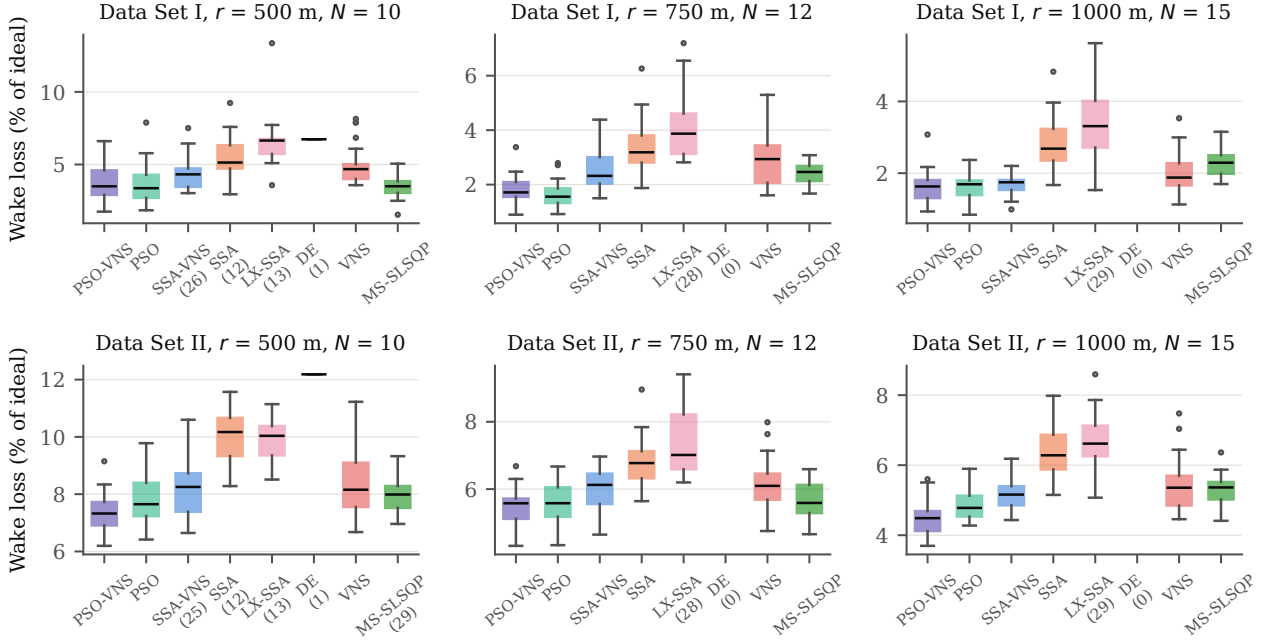


Figure S5: Distribution of the wake loss of the feasible runs (30 runs of 6,030 evaluations, random initialization) for the largest N of each farm (number of feasible runs in parentheses when fewer than 30).

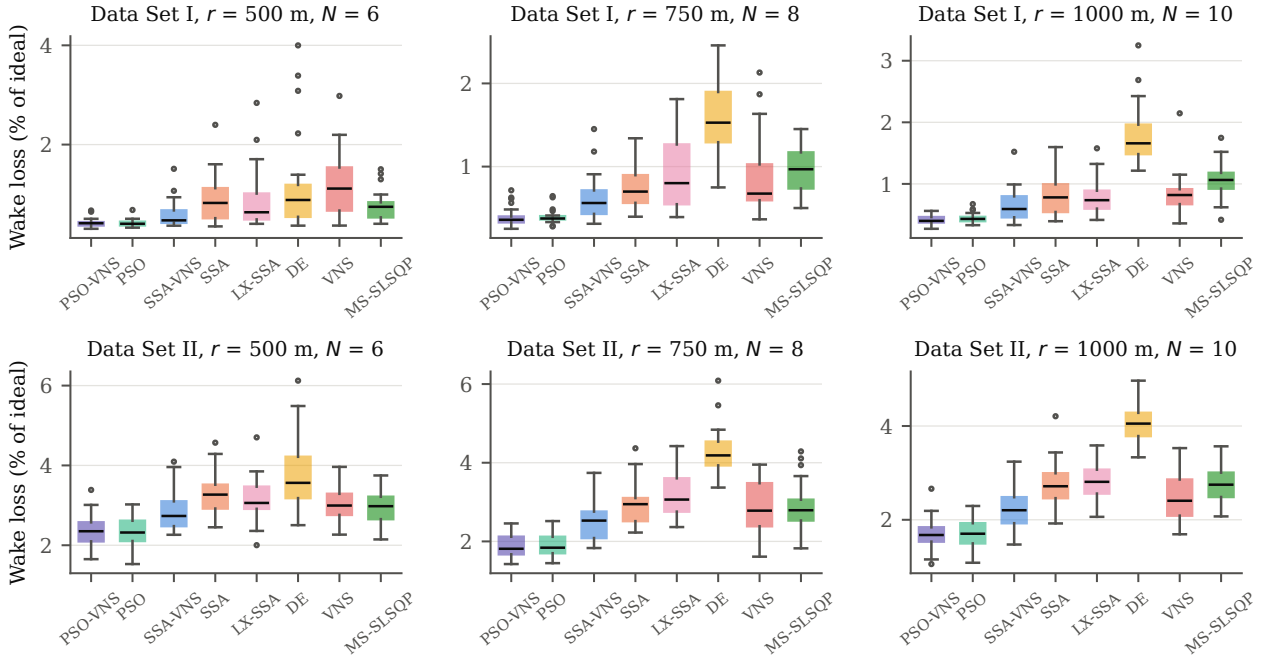


Figure S6: As Fig. S5, for the middle N of each farm.

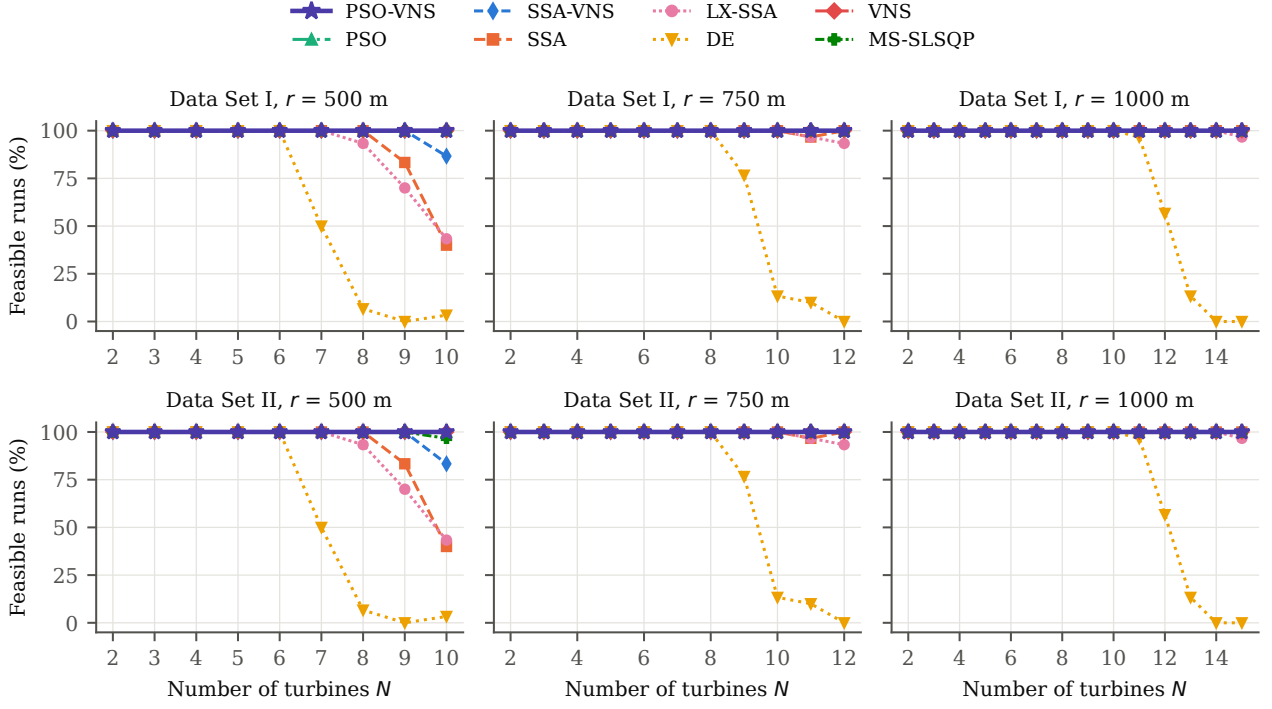


Figure S7: Percentage of feasible final layouts of the eight methods of the main comparison (30 seed-paired runs of 6,030 evaluations with random initialization per case) versus the number of turbines N , for each farm and wind data set.

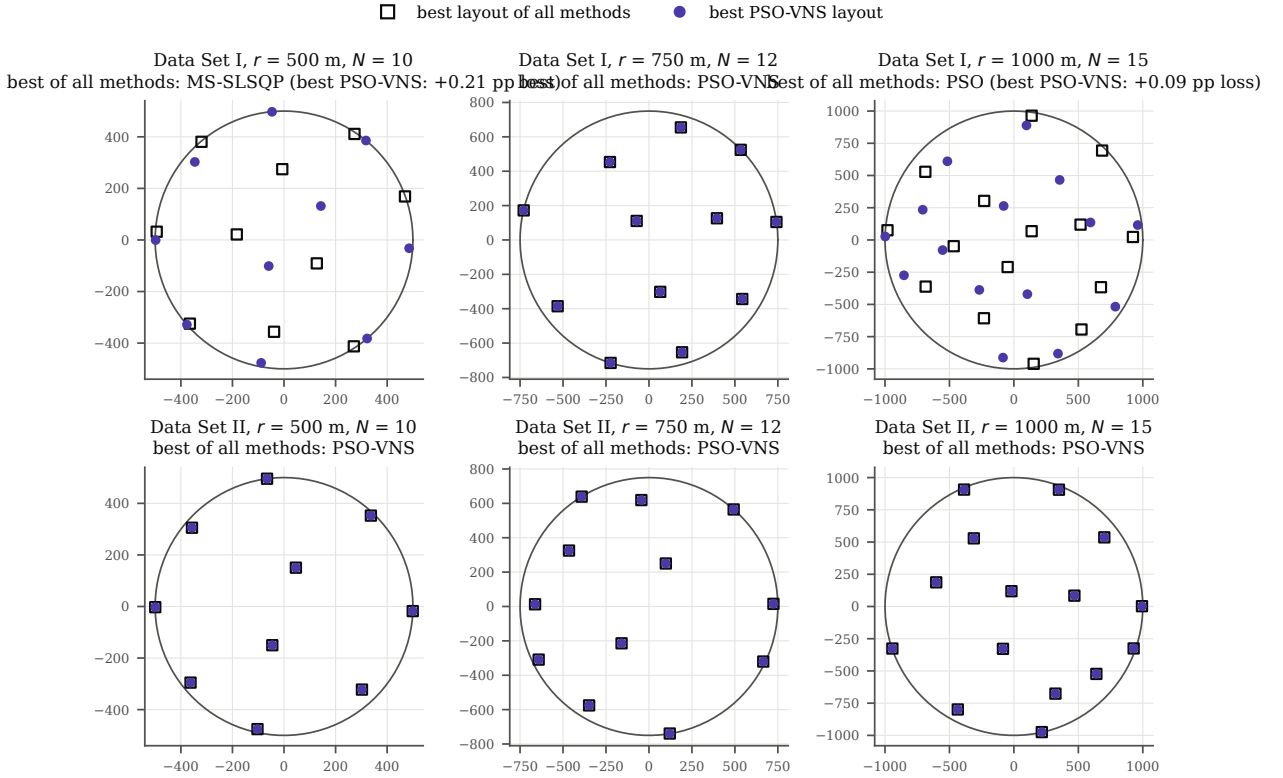


Figure S8: Best layout of all methods (squares) and best PSO-VNS layout (dots) for the largest N of each farm (one marker set when PSO-VNS is itself the best method); the coordinates are listed in Table S10.

S-III Component Analysis: Details

The component analysis of Section VII (Table IV) uses the same 68 cases and runs as the main comparison, plus RS-VNS and LX-SSA-VNS. RS-VNS is deliberately a weak, no-search control: its random layouts are uniform in the bounding square and are rarely feasible (Section S-I-D), so a Phase-1 method that is better than RS-VNS passes only a minimum bar. Against this control, an SSA phase gives only a small gain, an LX-SSA phase gives none, and the Laplace step makes the hybrid worse, whereas a PSO phase gives a clearly larger gain (Section VII). Table S11 gives the per-case values of LX-SSA-VNS and RS-VNS and the per-case outcomes of the contrasts of Table IV. Table S12 gives the feasibility and the wake loss of the Phase-1 result at the switch to VNS. Table S13 gives the per-case values of the budget-split study of Section VII, in which the split $\omega = 0.5$ of PSO-VNS, fixed before the experiments, is compared with $\omega = 0.25$ and $\omega = 0.75$ (the end point $\omega = 1$, plain PSO at the same budget, is included in Table V).

Table S11: Component analysis per case (68 cases, 6,030 evaluations, 30 seed-paired runs): mean wake loss (%) of the feasible runs of LX-SSA-VNS and RS-VNS (“ $\neg$ ”: fewer than 15 feasible runs), and run-level outcome of each planned contrast (+: first variant significantly better, -: significantly worse, $\cdot$: no significant difference; Wilcoxon signed-rank, Holm-adjusted over the 11 contrasts of the case). C1: PSO-VNS vs. PSO; C2: SSA-VNS vs. SSA; C3: LX-SSA-VNS vs. LX-SSA; C4: PSO-VNS vs. VNS; C5: PSO-VNS vs. RS-VNS; C6: SSA-VNS vs. RS-VNS; C7: LX-SSA-VNS vs. RS-VNS; C8: PSO-VNS vs. SSA-VNS; C9: PSO-VNS vs. LX-SSA-VNS; C10: SSA-VNS vs. LX-SSA-VNS; C11: LX-SSA vs. SSA.

DS	r	N	LX-SSA-VNS	RS-VNS	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11
I	500	2	0.000	0.000	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
I	500	3	0.033	0.027	$\cdot$	+	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
I	500	4	0.151	0.143	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
I	500	5	0.320	0.362	$\cdot$	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	500	6	0.770	1.168	$\cdot$	+	$\cdot$	+	+	+	+	+	+	$\cdot$	$\cdot$
I	500	7	1.692	1.955	$\cdot$	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	500	8	2.867	2.707	$\cdot$	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	500	9	3.586	4.198	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	500	10	4.888	4.879	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	750	2	0.000	0.000	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
I	750	3	0.000	-0.000	$\cdot$	+	+	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
I	750	4	0.031	0.032	$\cdot$	$\cdot$	+	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
I	750	5	0.114	0.103	$\cdot$	+	+	+	$\cdot$	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	750	6	0.241	0.218	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	750	7	0.446	0.462	$\cdot$	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	750	8	0.767	0.924	$\cdot$	$\cdot$	$\cdot$	+	+	+	$\cdot$	+	+	+	$\cdot$
I	750	9	0.989	1.215	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	750	10	1.706	1.661	$\cdot$	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	750	11	2.032	2.067	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	750	12	2.431	2.404	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	1000	2	0.000	0.000	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
I	1000	3	-0.000	-0.000	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
I	1000	4	0.006	0.008	$\cdot$	+	+	$\cdot$	+	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
I	1000	5	0.044	0.047	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	1000	6	0.118	0.104	$\cdot$	$\cdot$	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	1000	7	0.212	0.202	$\cdot$	+	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	1000	8	0.334	0.344	$\cdot$	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	1000	9	0.498	0.502	$\cdot$	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	1000	10	0.742	0.641	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	1000	11	1.004	0.983	$\cdot$	+	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	1000	12	1.220	1.011	$\cdot$	$\cdot$	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	1000	13	1.546	1.448	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	1000	14	1.640	1.671	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
I	1000	15	1.777	1.946	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	500	2	0.000	0.000	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
II	500	3	0.169	0.163	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
II	500	4	0.705	0.762	$\cdot$	$\cdot$	$\cdot$	$\cdot$	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	500	5	1.660	1.742	$\cdot$	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	500	6	2.861	3.024	$\cdot$	+	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	500	7	4.399	4.335	$\cdot$	$\cdot$	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	500	8	5.619	5.572	+	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	500	9	6.909	7.092	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	500	10	8.548	8.652	+	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	750	2	0.000	0.000	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
II	750	3	0.005	0.005	$\cdot$	+	+	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
II	750	4	0.133	0.178	$\cdot$	+	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
II	750	5	0.479	0.448	$\cdot$	+	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	750	6	0.909	0.960	$\cdot$	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	750	7	1.735	1.816	$\cdot$	$\cdot$	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	750	8	2.561	2.667	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	750	9	3.587	3.676	$\cdot$	+	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	750	10	4.603	4.380	$\cdot$	+	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	750	11	5.599	5.281	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	750	12	5.946	5.888	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	1000	2	0.000	0.000	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
II	1000	3	-0.000	-0.000	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
II	1000	4	0.054	0.037	$\cdot$	+	+	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
II	1000	5	0.180	0.182	$\cdot$	$\cdot$	+	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$
II	1000	6	0.446	0.409	$\cdot$	+	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	1000	7	0.775	0.683	$\cdot$	$\cdot$	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	1000	8	1.197	1.161	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	1000	9	1.836	1.641	$\cdot$	+	$\cdot$	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	1000	10	2.295	2.289	$\cdot$	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	1000	11	3.202	2.963	+	+	+	+	+	$\cdot$	$\cdot$	+	+	+	$\cdot$
II	1000	12	3.764	3.469	+	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	1000	13	4.276	4.281	+	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	1000	14	4.803	4.647	+	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$
II	1000	15	5.220	5.210	+	+	+	+	+	$\cdot$	$\cdot$	+	+	$\cdot$	$\cdot$

Table S12: Switch point of the two-phase variants (68 cases, 30 runs, 6,030 evaluations): first evaluation of the VNS phase, runs whose best layout is feasible at the switch (%), mean wake loss at the switch (%), feasible runs), mean and median share (%) of the wake loss left at the switch that the second phase removes, and runs infeasible at the switch that end feasible. “PSO continued”: PSO alone over the same evaluations.

Variant	Switch	Feas. (%)	Loss (%)	Mean share	Median share	Made feasible
PSO-VNS	3,030	98.7	1.63	30.3	22.6	26
SSA-VNS	3,030	95.7	2.01	36.7	29.2	79
LX-SSA-VNS	3,030	94.1	2.11	39.0	31.9	118
RS-VNS	3,030	66.7	1.30	57.1	54.3	668
PSO continued (no VNS)	3,030	98.7	1.63	29.2	20.6	26

Table S13: Sensitivity of PSO-VNS to the budget split between the PSO and VNS phases (25%, 50% and 75% of the 6,030 evaluations for PSO; 100%: PSO alone, same seeds): mean benchmark objective of the feasible runs (superscript: feasible runs when fewer than 30) and Wilcoxon signed-rank p (30 seed-paired runs), Holm-adjusted over the 4 comparisons of each case.

DS	r	N	$\omega = 0.25$	0.5	0.75	1	p (0.5/0.25)	p (0.5/0.75)	p (0.5/1)	p (0.75/1)
I	500	6	83851.3	83918.5	83929.5	83925.6	0.045	0.929	0.929	0.093
I	500	10	135131.6	135160.5	135421.8	135396.0	0.809	0.138	0.421	0.809
I	750	8	111835.7	111921.0	111944.0	111924.1	0.021	0.090	0.477	0.008
I	750	12	164894.0	165526.4	165991.2	165826.3	0.008	0.012	0.042	0.042
I	1000	10	139700.6	139874.2	139883.7	139834.3	0.002	0.529	0.001	1.49×10^{-8}
I	1000	15	207367.9	207308.1	207724.0	207263.9	0.741	6.39×10^{-5}	0.741	6.39×10^{-5}
II	500	6	42809.4	42851.5	42883.5	42857.9	0.165	0.192	0.984	2.24×10^{-4}
II	500	10	67624.1	67773.5	67799.8	67472.2	0.115	0.903	1.49×10^{-4}	1.77×10^{-5}
II	750	8	57358.0	57411.1	57448.4	57404.8	0.182	0.228	0.626	0.002
II	750	12	82784.6	82968.9	83182.0	82874.2	0.140	0.010	0.140	5.59×10^{-5}
II	1000	10	71769.3	71906.7	71976.0	71902.4	0.031	0.020	0.516	0.005
II	1000	15	104349.6	104794.8	104784.9	104375.6	4.63×10^{-4}	0.968	9.72×10^{-6}	2.46×10^{-7}

S-IV Tests Beyond the Benchmark: Details

This section gives the per-case values of the feasibility-preserving initialization and the budget scaling (Section VIII-A), the full results of the Horns Rev 1 block (Section VIII-B) and of IEA37 Case Study 1 (Section VIII-C), and the computational cost (Section VIII-D).

S-IV-A Feasibility-Preserving Initialization and Budget Scaling

Tables S14 and S15 give the per-case values behind Table VI for the six largest benchmark cases, together with the Horns Rev 1 block (the averages of Table VI are over the six benchmark cases only). In the paired tests of Table S14, runs are scored as in Section S-I-C: an infeasible run ranks below every feasible run, so a positive r_{rb} can come from more feasible runs even when the mean of the feasible runs is not better. With feasible starts, PSO and DE do not move away from the best initial layout (Section VIII-A), so their feasible-start values coincide in every case.

Table S14: Feasible versus random initialization per case (6,030 evaluations): mean wake loss (%; Horns Rev: mean AEP in GWh/yr) of the feasible runs, feasible runs, Holm-adjusted Wilcoxon signed-rank p (over the methods of the case) and rank-biserial r_{rb} (positive = feasible initialization better).

Method	Random	Feasible	Feas. (R)	Feas. (F)	p_{Holm}	r_{rb}
<i>Data Set I, $r = 500$ m, $N = 10$</i>						
PSO-VNS	3.771	2.871	30/30	30/30	0.008	+0.59
PSO	3.604	3.698	30/30	30/30	0.477	-0.15
SSA-VNS	4.369	2.932	26/30	30/30	2.76×10^{-6}	+0.91
SSA	5.582	3.611	12/30	30/30	9.31×10^{-8}	+0.98
LX-SSA	6.756	3.649	13/30	30/30	2.98×10^{-8}	+1.00
DE	6.737	3.698	1/30	30/30	1.68×10^{-8}	+1.00
VNS	4.905	2.782	30/30	30/30	2.98×10^{-8}	+1.00
MS-SLSQP	3.428	2.894	30/30	30/30	0.001	+0.70
LX-SSA-VNS	4.888	2.860	29/30	30/30	2.98×10^{-8}	+1.00
<i>Data Set I, $r = 750$ m, $N = 12$</i>						
PSO-VNS	1.793	1.909	30/30	30/30	0.339	-0.20
PSO	1.615	3.252	30/30	30/30	1.68×10^{-8}	-1.00
SSA-VNS	2.533	1.947	30/30	30/30	0.004	+0.66
SSA	3.349	2.623	30/30	30/30	0.006	+0.63
LX-SSA	4.105	2.695	28/30	30/30	8.34×10^{-6}	+0.90
DE	–	3.252	0/30	30/30	1.68×10^{-8}	+1.00
VNS	2.869	1.778	30/30	30/30	9.59×10^{-5}	+0.83
MS-SLSQP	2.415	2.169	30/30	30/30	0.110	+0.40
LX-SSA-VNS	2.431	1.911	30/30	30/30	0.001	+0.72
<i>Data Set I, $r = 1000$ m, $N = 15$</i>						
PSO-VNS	1.603	1.586	30/30	30/30	0.598	-0.11
PSO	1.624	3.162	30/30	30/30	1.68×10^{-8}	-1.00
SSA-VNS	1.693	1.561	30/30	30/30	0.328	+0.29
SSA	2.804	2.493	30/30	30/30	0.213	+0.42
LX-SSA	3.325	2.619	29/30	30/30	0.004	+0.68
DE	–	3.162	0/30	30/30	1.68×10^{-8}	+1.00
VNS	1.993	1.541	30/30	30/30	0.001	+0.74
MS-SLSQP	2.283	2.092	30/30	30/30	0.213	+0.42
LX-SSA-VNS	1.777	1.585	30/30	30/30	0.220	+0.38
<i>Data Set II, $r = 500$ m, $N = 10$</i>						
PSO-VNS	7.355	7.411	30/30	30/30	0.761	-0.07
PSO	7.767	9.147	30/30	30/30	1.89×10^{-6}	-0.93
SSA-VNS	8.266	7.401	25/30	30/30	1.49×10^{-4}	+0.79
SSA	9.942	8.831	12/30	30/30	3.26×10^{-7}	+0.97
LX-SSA	9.959	8.924	13/30	30/30	1.04×10^{-7}	+0.98
DE	12.185	9.147	1/30	30/30	1.68×10^{-8}	+1.00
VNS	8.336	7.263	30/30	30/30	2.76×10^{-6}	+0.91
MS-SLSQP	8.005	7.819	29/30	30/30	0.248	+0.32
LX-SSA-VNS	8.548	7.430	29/30	30/30	2.36×10^{-6}	+0.92
<i>Data Set II, $r = 750$ m, $N = 12$</i>						
PSO-VNS	5.486	5.304	30/30	30/30	0.192	+0.35
PSO	5.594	7.395	30/30	30/30	1.68×10^{-8}	-1.00
SSA-VNS	6.019	5.385	30/30	30/30	7.15×10^{-6}	+0.90
SSA	6.792	6.475	30/30	30/30	0.165	+0.40
LX-SSA	7.339	6.624	28/30	30/30	0.003	+0.68
DE	–	7.395	0/30	30/30	1.68×10^{-8}	+1.00
VNS	6.170	5.059	30/30	30/30	1.79×10^{-6}	+0.94
MS-SLSQP	5.687	5.646	30/30	30/30	0.919	-0.02
LX-SSA-VNS	5.946	5.365	30/30	30/30	1.73×10^{-4}	+0.80
<i>Data Set II, $r = 1000$ m, $N = 15$</i>						
PSO-VNS	4.498	4.549	30/30	30/30	0.556	-0.13
PSO	4.880	6.639	30/30	30/30	1.68×10^{-8}	-1.00
SSA-VNS	5.190	4.504	30/30	30/30	4.99×10^{-6}	+0.91
SSA	6.365	5.648	30/30	30/30	9.38×10^{-4}	+0.72
LX-SSA	6.747	5.775	29/30	30/30	1.53×10^{-6}	+0.94
DE	–	6.639	0/30	30/30	1.68×10^{-8}	+1.00
VNS	5.425	4.439	30/30	30/30	1.30×10^{-7}	+0.98
MS-SLSQP	5.313	5.145	30/30	30/30	0.192	+0.35
LX-SSA-VNS	5.220	4.559	30/30	30/30	1.99×10^{-4}	+0.79
<i>Horns Rev 1, 16 turbines</i>						
PSO-VNS	138.29	138.43	30/30	30/30	0.617	+0.22
PSO	137.79	136.39	19/30	30/30	0.617	+0.20
SSA-VNS	137.59	138.42	30/30	30/30	1.46×10^{-4}	+0.82
SSA	136.25	137.16	30/30	30/30	9.27×10^{-4}	+0.73
LX-SSA	136.00	137.07	28/30	30/30	1.66×10^{-4}	+0.82
DE	–	136.39	0/30	30/30	1.68×10^{-8}	+1.00
VNS	137.35	138.53	30/30	30/30	1.66×10^{-4}	+0.82
MS-SLSQP	136.31	136.90	30/30	30/30	0.007	+0.62
LX-SSA-VNS	137.41	138.36	30/30	30/30	1.66×10^{-4}	+0.81

Note: with feasible starts, PSO and DE do not move away from the best (packed) initial layout (Section VIII-A), so their feasible-start values coincide in every case. Infeasible runs enter the paired test ranked below every feasible run (Section S-I-C).

Table S15: Budget scaling per case: mean wake loss (%; Horns Rev: mean AEP in GWh/yr) of the feasible runs and feasible runs at each budget (random initialization).

Method	6,030	Feas.	30,030	Feas.	120,030	Feas.
<i>Data Set I, $r = 500$ m, $N = 10$ (wake loss, %)</i>						
PSO-VNS	3.771	30/30	2.506	30/30	2.343	30/30
PSO	3.604	30/30	2.951	30/30	2.890	30/30
SSA-VNS	4.369	26/30	2.978	30/30	2.305	30/30
SSA	5.582	12/30	4.799	28/30	3.948	30/30
LX-SSA	6.756	13/30	5.618	25/30	4.761	29/30
DE	6.737	1/30	3.111	30/30	3.077	30/30
VNS	4.905	30/30	3.078	30/30	2.142	30/30
MS-SLSQP	3.428	30/30	2.645	30/30	2.088	30/30
LX-SSA-VNS	4.888	29/30	3.416	30/30	2.494	30/30
RS-VNS	4.879	24/30	3.344	30/30	2.278	30/30
<i>Data Set I, $r = 750$ m, $N = 12$ (wake loss, %)</i>						
PSO-VNS	1.793	30/30	1.214	30/30	1.053	30/30
PSO	1.615	30/30	1.241	30/30	1.190	30/30
SSA-VNS	2.533	30/30	1.621	30/30	1.316	30/30
SSA	3.349	30/30	2.316	30/30	1.801	30/30
LX-SSA	4.105	28/30	2.556	30/30	2.203	30/30
DE	–	0/30	1.188	30/30	1.066	30/30
VNS	2.869	30/30	1.663	30/30	1.184	30/30
MS-SLSQP	2.415	30/30	1.872	30/30	1.543	30/30
LX-SSA-VNS	2.431	30/30	1.578	30/30	1.352	30/30
RS-VNS	2.404	30/30	1.711	30/30	1.339	30/30
<i>Data Set I, $r = 1000$ m, $N = 15$ (wake loss, %)</i>						
PSO-VNS	1.603	30/30	1.160	30/30	1.048	30/30
PSO	1.624	30/30	1.195	30/30	1.124	30/30
SSA-VNS	1.693	30/30	1.410	30/30	1.146	30/30
SSA	2.804	30/30	2.160	30/30	1.564	30/30
LX-SSA	3.325	29/30	2.489	30/30	2.018	30/30
DE	–	0/30	2.513	21/30	1.057	30/30
VNS	1.993	30/30	1.390	30/30	1.131	30/30
MS-SLSQP	2.283	30/30	1.844	30/30	1.608	30/30
LX-SSA-VNS	1.777	30/30	1.477	30/30	1.192	30/30
RS-VNS	1.946	30/30	1.496	30/30	1.175	30/30
<i>Data Set II, $r = 500$ m, $N = 10$ (wake loss, %)</i>						
PSO-VNS	7.355	30/30	6.750	30/30	6.434	30/30
PSO	7.767	30/30	6.979	30/30	6.861	30/30
SSA-VNS	8.266	25/30	7.097	30/30	6.488	30/30
SSA	9.942	12/30	8.300	28/30	7.759	30/30
LX-SSA	9.959	13/30	9.202	25/30	8.650	29/30
DE	12.185	1/30	6.874	30/30	6.789	30/30
VNS	8.336	30/30	7.011	30/30	6.461	30/30
MS-SLSQP	8.005	29/30	7.283	30/30	6.986	30/30
LX-SSA-VNS	8.548	29/30	7.101	30/30	6.561	30/30
RS-VNS	8.652	24/30	7.196	30/30	6.645	30/30
<i>Data Set II, $r = 750$ m, $N = 12$ (wake loss, %)</i>						
PSO-VNS	5.486	30/30	4.525	30/30	4.292	30/30
PSO	5.594	30/30	4.628	30/30	4.500	30/30
SSA-VNS	6.019	30/30	4.870	30/30	4.356	30/30
SSA	6.792	30/30	6.005	30/30	5.279	30/30
LX-SSA	7.339	28/30	6.486	30/30	5.952	30/30
DE	–	0/30	4.783	30/30	4.269	30/30
VNS	6.170	30/30	4.953	30/30	4.295	30/30
MS-SLSQP	5.687	30/30	5.173	30/30	4.837	30/30
LX-SSA-VNS	5.946	30/30	4.885	30/30	4.375	30/30
RS-VNS	5.888	30/30	4.919	30/30	4.410	30/30
<i>Data Set II, $r = 1000$ m, $N = 15$ (wake loss, %)</i>						
PSO-VNS	4.498	30/30	3.862	30/30	3.699	30/30
PSO	4.880	30/30	4.079	30/30	3.951	30/30
SSA-VNS	5.190	30/30	4.312	30/30	3.754	30/30
SSA	6.365	30/30	5.276	30/30	4.809	30/30
LX-SSA	6.747	29/30	6.091	30/30	5.740	30/30
DE	–	0/30	5.671	21/30	3.584	30/30
VNS	5.425	30/30	4.326	30/30	3.832	30/30
MS-SLSQP	5.313	30/30	4.848	30/30	4.440	30/30
LX-SSA-VNS	5.220	30/30	4.616	30/30	3.860	30/30
RS-VNS	5.210	30/30	4.510	30/30	3.681	30/30
<i>Horns Rev 1, 16 turbines (AEP, GWh/yr)</i>						
PSO-VNS	138.29	30/30	139.11	30/30	139.29	10/10
PSO	137.79	19/30	138.72	28/30	138.94	10/10
SSA-VNS	137.59	30/30	138.11	30/30	139.17	10/10
SSA	136.25	30/30	136.65	30/30	137.47	10/10
LX-SSA	136.00	28/30	136.16	30/30	136.48	10/10
DE	–	0/30	137.79	8/30	139.70	10/10
VNS	137.35	30/30	138.61	30/30	139.54	10/10
MS-SLSQP	136.31	30/30	137.18	30/30	137.93	10/10
LX-SSA-VNS	137.41	30/30	138.28	30/30	139.22	10/10
RS-VNS	137.58	28/30	138.34	30/30	139.51	10/10

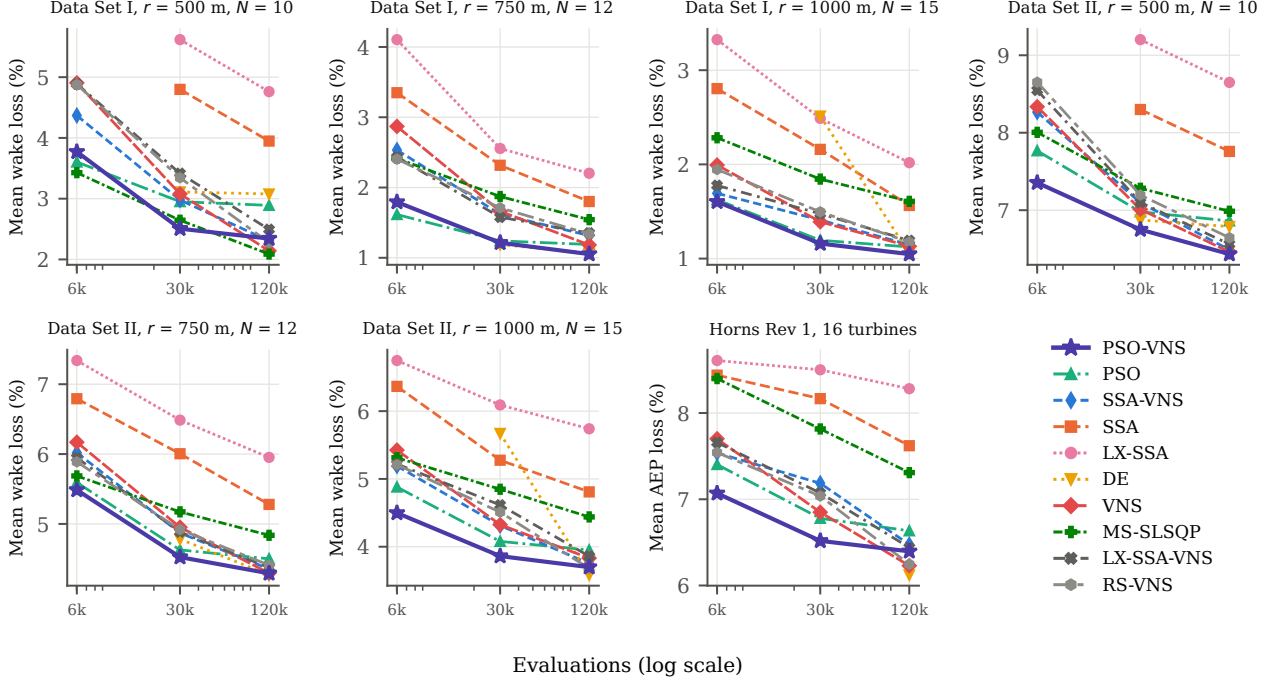


Figure S9: Budget scaling: mean wake loss of the six largest benchmark cases and mean AEP loss of the Horns Rev 1 block (feasible runs; methods with at least half of their runs feasible) versus the budget (6,030, 30,030 and 120,030 evaluations, random initialization; 10 seeds for Horns Rev 1 at 120,030 evaluations).

S-IV-B Horns Rev 1 16-Turbine Block

Table S16 gives the full results for the Horns Rev 1 block, and Fig. S10 its convergence and layouts. All values come from the runs with the corrected direction binning (Section S-I-G). *Check against PyWake*. Our model was checked on the installed 80-turbine Horns Rev 1 farm against PyWake’s Jensen (NOJ) model [9] with the same turbine curves, wind climate and wake-expansion constant $K = 0.04$: the AEP values agree within 0.64%. The remaining difference is expected from the simplifications of our model (thrust coefficient at the free-stream speed, hub-centre in-wake test instead of a rotor-averaged deficit).

Table S16: Horns Rev 1 site case, 16-turbine block (Vestas V80 power and thrust curves, measured 12-sector wind climate, installed outline; Jensen wake $k = 0.04$; minimum spacing $4D = 320$ m; wake-free AEP 148.81 GWh/yr): AEP in GWh/yr, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 evaluations, mean wake loss (%), feasible runs, Wilcoxon signed-rank p of PSO-VNS vs. each method (Holm-adjusted over these 9 comparisons) and rank-biserial r_{rb} (positive = PSO-VNS better; zero differences dropped); run-level Friedman $p = 1.64 \times 10^{-29}$ (over the seeds common to all methods).

Layout / method	AEP	Loss (%)	Feas.	p_{Holm}	r_{rb}
Installed layout	139.82	6.04	—	—	—
PSO-VNS	138.29 (0.75)	7.07	30/30	—	—
SSA-VNS	137.59 (0.72)	7.54	30/30	2.56×10^{-4}	+0.72
LX-SSA-VNS	137.41 (0.85)	7.66	30/30	1.58×10^{-4}	+0.77
RS-VNS	137.58 (0.67)	7.54	28/30	1.08×10^{-4}	+0.81
VNS	137.35 (1.04)	7.70	30/30	1.08×10^{-4}	+0.81
PSO	137.79 (0.88)	7.40	19/30	4.61×10^{-5}	+0.85
SSA	136.25 (1.01)	8.44	30/30	2.12×10^{-7}	+0.97
LX-SSA	136.00 (0.83)	8.61	28/30	6.52×10^{-8}	+0.99
DE	—	—	0/30	1.68×10^{-8}	+1.00
MS-SLSQP	136.31 (0.83)	8.40	30/30	1.68×10^{-8}	+1.00

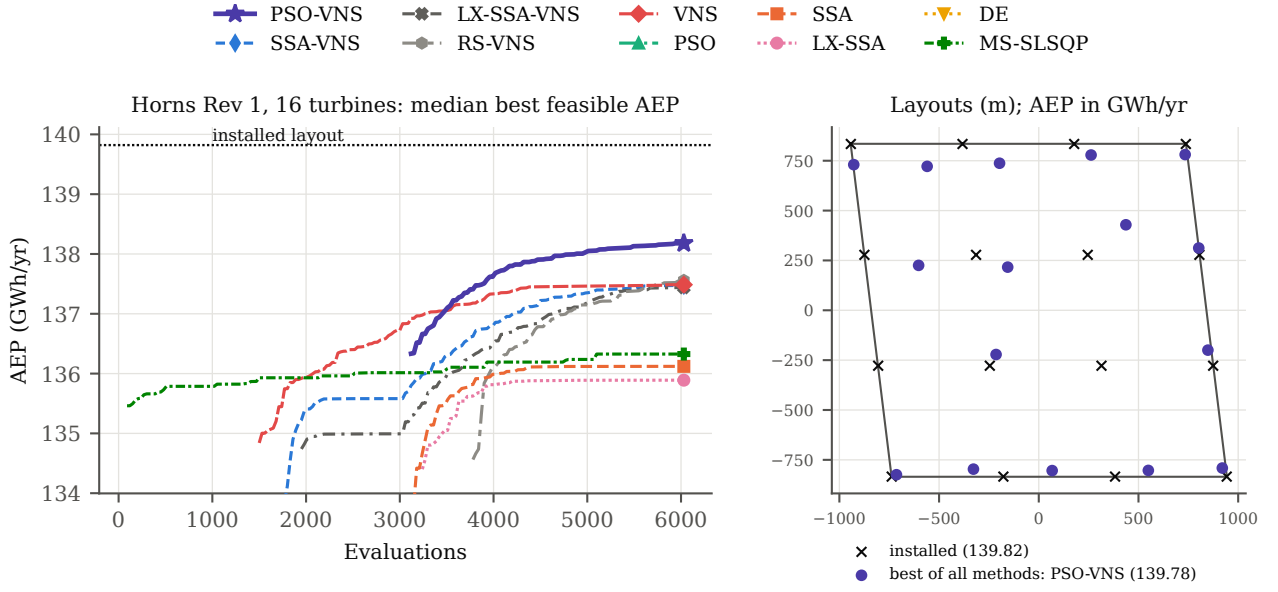


Figure S10: Horns Rev 1 16-turbine block (corrected direction binning): median best feasible AEP versus evaluations (left) and layouts (right).

S-IV-C IEA37 Case Study 1

Table S17 gives the results of all methods at both budgets, Table S18 compares our best layouts with the submitted ones [12, 13], Fig. S11 shows the convergence at the largest budget, and Fig. S12 the best layouts of IEA37 Case Study 1 and of the Horns Rev 1 block.

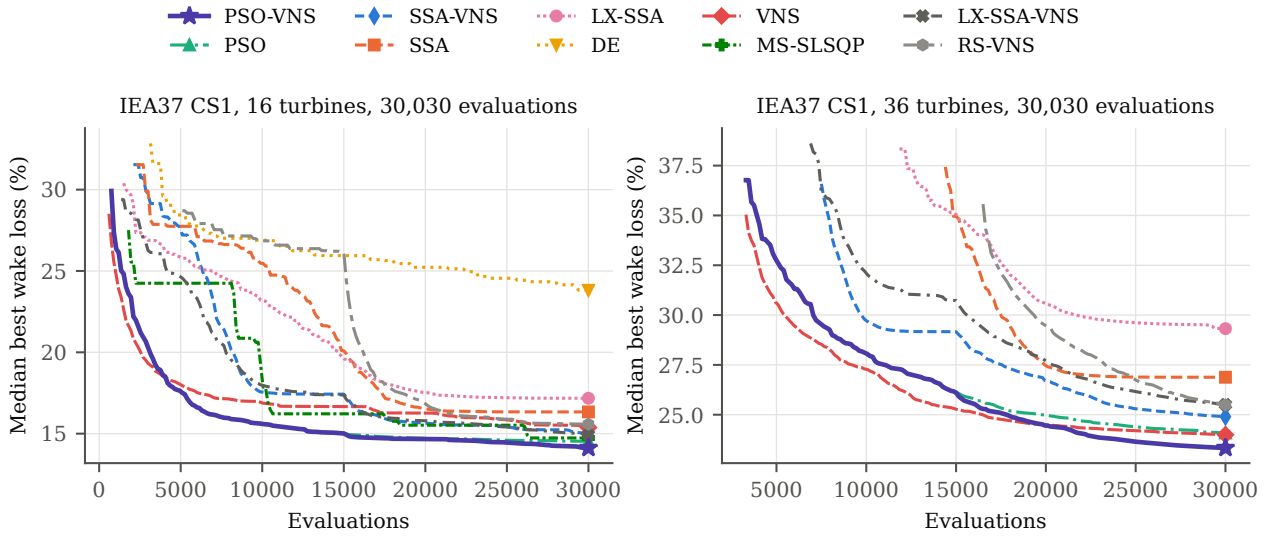


Figure S11: IEA37 Case Study 1: median best feasible wake loss versus evaluations at the largest budget.

Table S17: IEA Wind Task 37 Case Study 1 (16- and 36-turbine scenarios; official AEP model): AEP (MWh) mean, best and SD over the feasible runs, mean wake loss relative to the wake-free AEP (%), feasible runs, rank (ranking rule of Table II) and Holm-adjusted Wilcoxon signed-rank p of PSO-VNS vs. each method, per scenario and budget.

Method	Mean	Best	SD	Loss (%)	Feas.	Rank	p_{Holm}
<i>IEA37 CS1, 16 turbines, 6,030 evaluations</i>							
PSO-VNS	395,032	402,782	4,587	15.87	30/30	1	–
PSO	390,540	403,753	7,192	16.82	30/30	2	7.15×10^{-6}
SSA-VNS	388,563	403,251	8,564	17.25	30/30	3	3.80×10^{-4}
VNS	387,637	400,169	5,591	17.44	30/30	4	4.61×10^{-5}
LX-SSA-VNS	386,930	397,429	6,911	17.59	30/30	5	4.61×10^{-5}
RS-VNS	386,206	398,682	5,817	17.75	30/30	6	4.61×10^{-5}
MS-SLSQP	382,605	405,434	20,279	18.51	18/30	7	6.91×10^{-5}
SSA	378,073	395,150	9,076	19.48	30/30	8	6.52×10^{-8}
LX-SSA	373,977	393,285	7,867	20.35	30/30	9	1.68×10^{-8}
DE	338,611	357,391	8,837	27.88	27/30	10	1.68×10^{-8}
<i>IEA37 CS1, 16 turbines, 30,030 evaluations</i>							
PSO-VNS	402,583	413,332	4,678	14.26	30/30	1	–
PSO	401,347	412,702	4,764	14.52	30/30	2	0.006
LX-SSA-VNS	399,197	407,330	4,330	14.98	30/30	3	0.006
SSA-VNS	396,921	406,925	6,267	15.47	30/30	4	4.42×10^{-4}
VNS	396,192	404,076	4,928	15.62	30/30	5	7.99×10^{-5}
RS-VNS	396,163	408,205	4,977	15.63	30/30	6	8.52×10^{-6}
MS-SLSQP	394,567	410,350	15,316	15.97	29/30	7	0.006
SSA	392,876	401,459	4,394	16.33	30/30	8	4.30×10^{-7}
LX-SSA	390,267	398,770	5,052	16.88	30/30	9	1.68×10^{-8}
DE	359,952	397,403	12,352	23.34	30/30	10	2.98×10^{-8}
<i>IEA37 CS1, 36 turbines, 6,030 evaluations</i>							
VNS	741,566	785,791	24,100	29.81	30/30	1	0.404
PSO-VNS	736,832	787,709	19,268	30.25	30/30	2	–
PSO	729,189	787,989	21,150	30.98	27/30	3	8.49×10^{-4}
SSA-VNS	728,870	766,803	22,859	31.01	30/30	4	0.248
SSA	717,258	750,707	19,300	32.11	30/30	5	0.001
LX-SSA-VNS	715,952	757,838	30,626	32.23	29/30	6	0.002
LX-SSA	709,661	746,301	19,845	32.83	30/30	7	4.75×10^{-4}
RS-VNS	709,276	750,170	21,667	32.86	28/30	8	3.60×10^{-5}
MS-SLSQP	733,342	748,571	13,798	30.58	4/30	9	1.49×10^{-7}
DE	–	–	–	–	0/30	10	1.68×10^{-8}
<i>IEA37 CS1, 36 turbines, 30,030 evaluations</i>							
PSO-VNS	807,446	827,385	11,697	23.57	30/30	1	–
PSO	802,252	830,256	13,260	24.06	30/30	2	4.87×10^{-5}
VNS	801,797	824,230	11,876	24.10	30/30	3	0.119
SSA-VNS	791,404	819,931	14,356	25.09	30/30	4	4.18×10^{-4}
RS-VNS	787,892	807,208	9,532	25.42	30/30	5	4.16×10^{-6}
LX-SSA-VNS	785,677	810,989	13,216	25.63	30/30	6	1.12×10^{-7}
SSA	771,298	802,538	14,756	26.99	30/30	7	2.98×10^{-8}
LX-SSA	752,030	789,311	18,838	28.82	30/30	8	2.98×10^{-8}
MS-SLSQP	798,913	833,365	41,511	24.38	14/30	9	2.37×10^{-4}
DE	–	–	–	–	0/30	10	1.68×10^{-8}

Table S18: IEA Wind Task 37 Case Study 1: AEP (MWh, official calculator) of the baseline (example) layout, of the best feasible and the best overall participant layout, our best feasible layout over all methods and budgets, the best PSO-VNS layout, mean PSO-VNS AEP (budgets 6,030 / 30,030 evaluations), and difference (%) of the best PSO-VNS layout from the best feasible participant layout and from the baseline. “infeas.”: violates the boundary or spacing constraint by more than 1 mm.

Scenario	Baseline	Best feasible publ.	Best publ.	Our best	PSO-VNS best	PSO-VNS mean	Δ_{publ}	Δ_{base}
IEA37 CS1, 16 turbines	366,941.6	418,924.4 (par4)	421,561.9 (par12, infeas.)	413,331.9 (PSO-VNS)	413,331.9	395,031.8 / 402,583.1	-1.33	+12.64
IEA37 CS1, 36 turbines	737,883.1	863,676.3 (par4)	882,383.3 (par12, infeas.)	833,365.0 (MS-SLSQP)	827,385.0	736,832.4 / 807,446.0	-4.20	+12.13

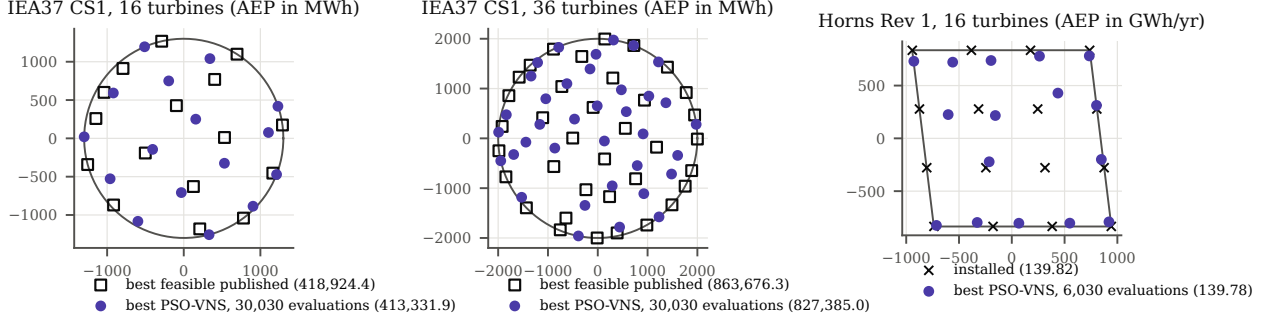


Figure S12: Best layouts. Left and middle: IEA37 Case Study 1, 16 and 36 turbines, best feasible published layout and best PSO-VNS layout at 30,030 evaluations. Right: Horns Rev 1 16-turbine block (corrected direction binning), installed layout (7D grid) and best PSO-VNS layout at 6,030 evaluations (minimum spacing 4D).

S-IV-D Computational Cost

Table S19 gives the time per evaluation and per run. The metaheuristics need 0.23–0.45 ms per evaluation and MS-SLSQP 0.53 ms, about 2.3 times as much as PSO. These timings are indicative only: they come from the machines that produced the respective result files, so small differences between methods are not meaningful.

Table S19: Computational cost at 6,030 evaluations: mean wall-clock time per evaluation (including the optimizer overhead) and range over N of the mean time per run (single core). Times come from the machine that produced each result file.

Method	ms per evaluation	s per run
PSO-VNS	0.349	1.5–3.2
PSO	0.244	1.0–2.2
SSA-VNS	0.388	1.7–3.4
SSA	0.399	1.7–3.6
LX-SSA	0.396	1.7–3.6
DE	0.467	2.0–4.3
VNS	0.371	1.6–3.4
MS-SLSQP	1.820	2.6–21.7

S-V Sensitivity to the Modeling Assumptions: Details

This section defines the Gaussian wake model of the re-evaluation in Section IX and gives the full re-evaluation results and the details of the spacing and boundary checks of Section IX.

S-V-A Gaussian Wake Model Used for the Re-Evaluation

For the re-evaluation of Section IX, the deficit of the Jensen model is replaced by the Gaussian wake model of Bastankhah and Porté-Agel [10],

$$\Delta_{ij} = \left(1 - \sqrt{1 - C_T/(8\sigma^2/D^2)}\right) \exp(-v_{ij}^2/(2\sigma^2)), \quad \sigma/D = K_G d_{ij}/D + \epsilon, \quad (S2)$$

$$\epsilon = 0.2\sqrt{\frac{1}{2}(1 + \sqrt{1 - C_T})/\sqrt{1 - C_T}},$$

for turbines i downstream of j , with $K_G = 0.04$, the lateral offset v_{ij} and downstream distance d_{ij} of Section II, and the same root-sum-square superposition and Weibull scaling as the benchmark. The Gaussian model differs from the Jensen cone both in its smooth lateral profile and in its expansion rate ($K_G = 0.04$ vs. $K = 0.075$, not calibrated to each other), and, unlike the cone test of the benchmark, it gives no deficit to turbines upstream of j . It is not valid in the near wake (below about 2–3 rotor diameters), which does not matter at $\ell_{\min} = 4D$.

S-V-B Re-Evaluation, Spacing and Boundary Handling

Table S20 gives the full re-evaluation of the final layouts under the alternative power-curve and wake models, and Table S21 gives the constructive capacity of the three farms at 4D, 5D and 6D spacing. Table S22 is an archived check from the previous version of this study: six cases were re-optimized at 5D and 6D with the original SSA and LX-SSA code only, and PSO-VNS and PSO were not re-run at larger spacing. It shows how

the attainable benchmark objective changes when the spacing is enlarged; it is not a comparison of the methods of this paper.

Table S20: Robustness of the results to the benchmark model. All feasible final layouts of the 8 methods are re-evaluated (not re-optimized) with alternative power-curve and wake models. Δ : mean relative change of the objective (expected power) with respect to the benchmark model (both evaluated on the stored coordinates, rounded to 1 mm); average rank of each method over the 68 cases (ranking rule of Table II: methods with fewer than half of their runs feasible are ranked last, by their number of feasible runs); $\bar{\tau}$: mean Kendall rank correlation between the benchmark and the alternative ordering of the methods with at least 15 feasible runs within a case; “Same best”: percentage of cases in which the best method is unchanged.

Model	Δ (%)		Average rank								$\bar{\tau}$	Same best (%)
	DS I	DS II	DSO-VNS	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP		
Benchmark (linear curve, Jensen)	+0.0	+0.0	1.82	1.97	3.45	5.38	6.06	7.04	4.40	5.88	–	–
Cubic power curve	-17.7	-36.9	1.82	1.90	3.53	5.68	5.97	6.90	4.42	5.79	0.95	94
Cubic curve with 25 m/s cut-out	-21.5	-37.0	1.80	1.91	3.53	5.68	5.97	6.90	4.42	5.79	0.95	93
Gaussian wake ($k^* = 0.04$)	-0.3	-0.4	1.97	1.91	3.93	5.43	6.04	6.94	5.12	4.66	0.66	54

Table S21: Largest turbine count for which a layout satisfying the circular boundary and the minimum spacing was constructed (multi-start packing search; constructive lower bound), compared with the largest tested N .

r (m)	4D (308 m)	5D (385 m)	6D (462 m)	Largest tested N
500	13	8	7	10
750	26	19	13	12
1000	42	29	21	15

Table S22: Minimum-spacing re-optimization with the original LX-SSA and SSA code (6,030 evaluations, 30 seed-paired runs per cell): mean benchmark objective, Wilcoxon signed-rank p for LX-SSA vs. SSA, and the mean LX-SSA wake loss as a percentage of the ideal objective at 4D and 6D. A superscript gives the number of feasible runs when fewer than 30.

DS	r (m)	N	4D (308 m)			5D (385 m)			6D (462 m)			LX-SSA loss (%)
			LX-SSA	SSA	p	LX-SSA	SSA	p	LX-SSA	SSA	p	
1	500	4	56078.8	56086.8	0.237	56082.4	56085.2	0.700	56075.5	56083.9	0.839	0.19 → 0.19
1	750	8	111374.0	111506.5	0.245	111154.1	111163.4	0.871	110314.4 ²⁸	110847.5	0.021	0.88 → 1.85
1	1000	8	111924.6	111947.1	0.887	111953.5	111946.4	0.777	111781.5	111964.7	0.005	0.39 → 0.52
2	500	4	29004.2	29028.7	0.146	28999.5	29019.6	0.229	28981.8	28971.1	0.612	0.88 → 0.96
2	750	8	56659.2	56806.2	0.061	56621.0	56673.3	0.584	56533.9 ²⁸	56690.6	0.038	3.18 → 3.47
2	1000	8	57645.1	57739.3	0.084	57580.7	57655.7	0.140	57475.5	57587.2	0.114	1.50 → 1.79

S-VI Model Derivations

Figs. S13–S15 illustrate the farm geometry and the wake model of Section II, and Section S-VI-A gives the discrete form of the expected power used by the benchmark objective.

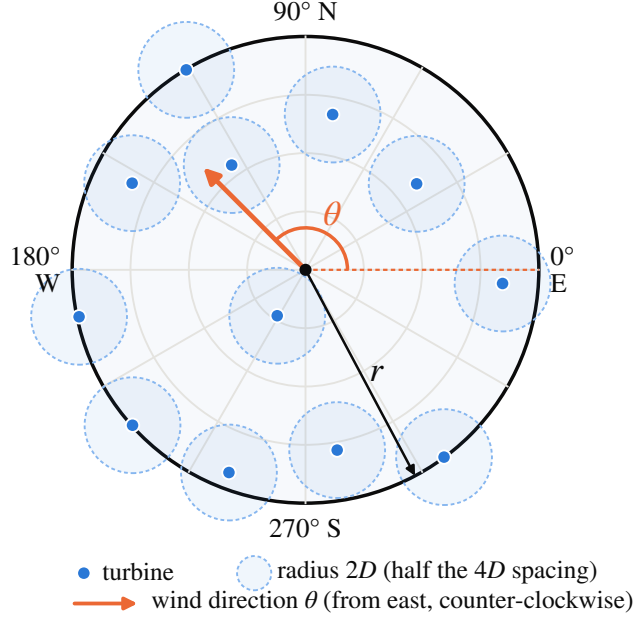


Figure S13: Circular wind farm of radius r . The wind direction θ (direction the wind blows towards) is measured counter-clockwise from east. The points show, for illustration only, the best LX-SSA-VNS layout for 12 turbines in the 750-m farm (Wind Data Set I) from the previous version of this study; the shaded discs have radius $2D$, so two discs touch when their turbines are exactly $4D$ apart.

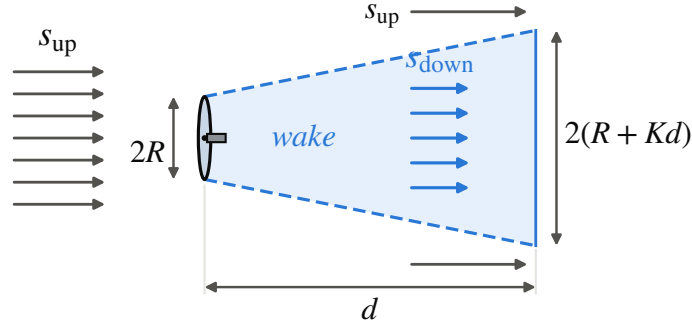


Figure S14: Jensen wake behind a rotor of radius R (schematic, not to scale). The wake widens linearly with the spreading constant K , so that its width at the downstream distance d is $2(R + Kd)$, and the wind speed inside it drops from s_{up} to s_{down} .

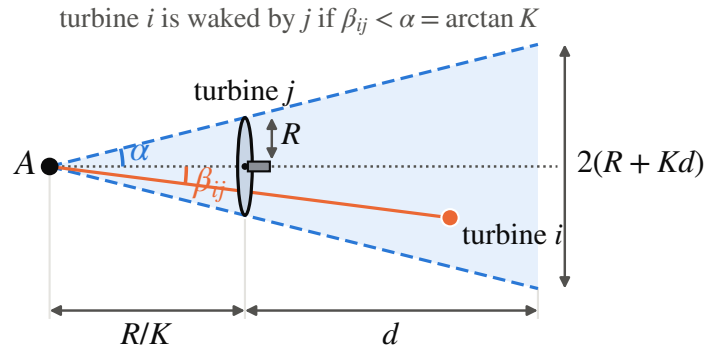


Figure S15: Wake half cone of turbine j (schematic, K exaggerated). The virtual vertex A lies a distance R/K upstream of the rotor, the half-cone angle is $\alpha = \arctan K$, and a turbine i lies in the wake of j when $\beta_{ij} < \alpha$. As in [1], the test also admits a turbine that lies less than R/K upstream of j and close to its axis (between A and the rotor); this non-physical upstream deficit is kept for comparability with the published results.

S-VI-A Discrete Form of the Expected Power

Following [1], the integrals of (3) are evaluated with a Riemann sum. The wind direction is divided into bins $\theta_0 = 0^\circ, \theta_1, \dots, \theta_{N_\theta} = 360^\circ$ with probabilities p_l , and the speed range between $s_{\text{cut-in}}$ and s_{rated} into bins $s_0 = s_{\text{cut-in}}, s_1, \dots, s_{N_s} = s_{\text{rated}}$. With $\bar{\theta}_l = (\theta_{l-1} + \theta_l)/2$ and $W_i(s, \theta) = \exp\{-[s/\psi_i(\theta)]^{\zeta(\theta)}\}$,

$$E(P_i) = \sum_{l=1}^{N_\theta} (\theta_l - \theta_{l-1}) p_l \left\{ P_{\text{rated}} W_i(s_{\text{rated}}, \bar{\theta}_l) + \lambda_1 \sum_{q=1}^{N_s} \frac{s_{q-1} + s_q}{2} [W_i(s_{q-1}, \bar{\theta}_l) - W_i(s_q, \bar{\theta}_l)] \right. \\ \left. + \lambda_0 [W_i(s_{\text{cut-in}}, \bar{\theta}_l) - W_i(s_{\text{rated}}, \bar{\theta}_l)] \right\}. \quad (\text{S3})$$

Like the earlier studies, we keep the bin-width factor $(\theta_l - \theta_{l-1})$. Because all 24 direction bins are 15° wide and $\sum_l p_l = 1$, the objective equals 15 times the expected farm power in kW, and the gross (wake-only) AEP of the benchmark is $\text{AEP [MWh]} = (\text{objective}/15) \times 8.76$. The linear part of the power curve is -7.0 kW at 3.5 m/s and zero at 3.55 m/s.

S-VII Published Kusiak–Song Results

The Kusiak–Song benchmark has been solved with an evolution strategy [1], ant colony optimization [2], particle filtering [3] and biogeography-based optimization [4]. We do not compare our results numerically with these studies. The published values available to us are transcriptions whose source tables could not be checked against the original papers, several of them are internally inconsistent (e.g., objective and wake loss that do not add up to the wake-free value), and the budgets, the numbers of runs and the handling of infeasible layouts of these studies differ from ours or are not reported. The transcribed values and the status of each are listed in `analysis/literature_values.csv` and `analysis/literature_values.md`.

S-VIII Reproducibility Map

Table S23 lists which script produces which part of the results; `analysis/README_reproduce.md` gives the commands, the run files and the order in which to call them. The per-run results of the main comparison come from two generations of experiment scripts with the same objective and optimizer code: SSA, LX-SSA, DE, the old-setting PSO, VNS, SSA-VNS and LX-SSA-VNS from the grid scripts of the previous version (`fresh_*.csv`), and PSO, PSO-VNS, RS-VNS and MS-SLSQP, as well as the split, initialization, budget, Horns Rev 1 and IEA37 studies, from `analysis/mpce_experiments.py` and `analysis/iea37_experiments.py` (`mpce_*.csv`); `mpce_results.py` records the source file of every run. Later code changes were behavior-preserving: re-runs of archived runs with the current code reproduce their objectives and coordinates exactly. Convergence curves are stored at every $\lfloor (B - 30)/200 \rfloor$ evaluations. Table S24 gives the source of the runs of every method and study.

Table S23: Reproducibility map: script (in `analysis/`) that produces each part of the results. Commands and run files: `analysis/README_reproduce.md`.

Results	Script
Runs of PSO, PSO-VNS, RS-VNS, MS-SLSQP; split, feasible-initialization, budget and Horns Rev 1 studies	<code>mpce_experiments.py</code> (one experiment id per study)
Runs of IEA37 Case Study 1	<code>iea37_experiments.py</code>
Runs of SSA, LX-SSA, DE, old-setting PSO, VNS, SSA-VNS, LX-SSA-VNS	grid scripts of the previous version (<code>full_grid_experiments.py</code> and successors)
Main-text tables, generated tables of this document (<code>mpce_supplementary.tex</code>), figures in <code>figures/mpce/</code> , <code>mpce_summary.json</code>	<code>mpce_results.py</code> (called by <code>build_mpce_paper.sh</code>)
Number macros <code>mpce_numbers.tex</code> ; checks of outcome-dependent statements	<code>mpce_numbers.py</code> ; <code>mpce_check_final.py</code> (both called by <code>mpce_results.py</code>)
Re-evaluation with the cubic power curve and the Gaussian wake	<code>mpce_robustness.py</code> (cached in <code>mpce_reevaluation.cache.csv</code>)
Spacing re-optimization with SSA and LX-SSA (Table S22)	<code>analyze_authors_runs.py</code>
Packing capacity (Table S21)	<code>packing_capacity.py</code>
Model schematics (Figs. S13–S15)	<code>make_model_figures.py</code>
Verification of the evaluator against the original code and archived runs	<code>validate_evaluator.py</code> , <code>calibrate_authors_code.py</code>

Table S24: Provenance of the per-run results used in this paper: study, budget (evaluations), initialization, methods, source file (mpce_<exp>: all shards mpce_<exp>_s<i>of<k>.csv of experiment <exp> of mpce_experiments.py / iea37_experiments.py; fresh_*.csv: earlier grid runs, see the repository README) and number of runs.

Study	Budget	Init.	Methods	Source	Runs
Benchmark	6,030	feasible	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, MS-SLSQP, SSA, SSA-VNS	mpce_feas	1440
Benchmark	6,030	feasible	PSO-VNS	mpce_feasp	180
Benchmark	6,030	random	LX-SSA-VNS, SSA-VNS	fresh_bgrid.csv	4080
Benchmark	6,030	random	DE, LX-SSA, PSO (old), SSA	fresh_grid.csv	8160
Benchmark	6,030	random	VNS	fresh_vgrid.csv	2040
Benchmark	6,030	random	PSO-VNS	mpce_psohv	2040
Benchmark	6,030	random	PSO	mpce_pso	2040
Benchmark	6,030	random	PSO-VNS ($\omega = 0.25$), PSO-VNS ($\omega = 0.75$)	mpce_pso	720
Benchmark	6,030	random	RS-VNS	mpce_rsvns	2040
Benchmark	6,030	random	MS-SLSQP	mpce_slsqp	2040
Benchmark	30,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_b30k	1620
Benchmark	30,030	random	PSO-VNS	mpce_b30kp	180
Benchmark	120,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_b120k	1620
Benchmark	120,030	random	PSO-VNS	mpce_b120kp	180
Horns Rev 1	6,030	feasible	VNS, DE, LX-SSA-VNS, LX-SSA, PSO-VNS, PSO, MS-SLSQP, SSA, SSA-VNS	mpce_hrfix	270
Horns Rev 1	6,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO-VNS, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_hrfix	300
Horns Rev 1	30,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO-VNS, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_hrfix	300
Horns Rev 1	120,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO-VNS, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_hrfix	100
IEA37	6,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_iea16	270
IEA37	6,030	random	PSO-VNS	mpce_iea16p	30
IEA37	6,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_iea36	270
IEA37	6,030	random	PSO-VNS	mpce_iea36p	30
IEA37	30,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_iea16	270
IEA37	30,030	random	PSO-VNS	mpce_iea16p	30
IEA37	30,030	random	VNS, DE, LX-SSA-VNS, LX-SSA, PSO, RS-VNS, MS-SLSQP, SSA, SSA-VNS	mpce_iea36	270
IEA37	30,030	random	PSO-VNS	mpce_iea36p	30

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